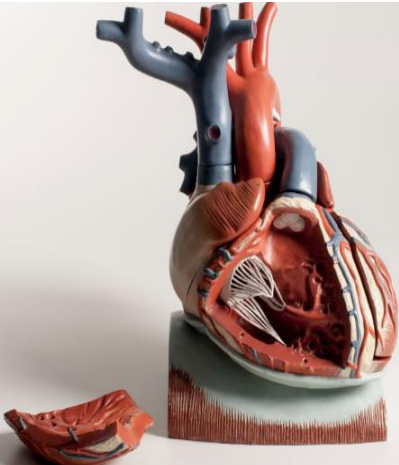
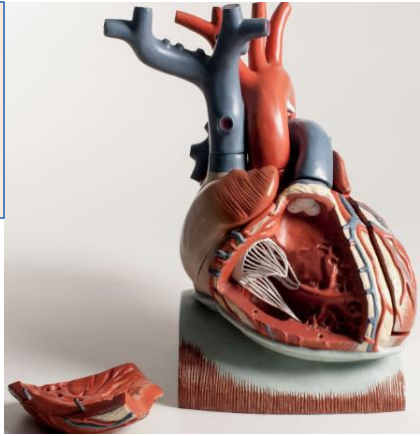




Biochemistry of CVS



For PC-I

By:

Abebe D. Dubiwak (BSc, MSc, PhD Candidate)

May, 2026

Jimma, Ethiopia

Session objectives

- At the end of the session student will able to:
 - ✓ Describe Cardiac energetics
 - ✓ Discuss mechanism of atherosclerosis
 - ✓ Distinguish biomarkers of cardiac injury

Cardiac metabolism



- **Cardiac muscle** is adapted to be highly resistant to fatigue:
 - ✓ Large number of myoglobin & mitochondria → continuous aerobic ATP production via oxidative phosphorylation.
- The heart is so tuned to **aerobic metabolism** that it is unable to pump sufficiently in ischemic conditions.
- Normally, about 1% of energy is derived from **anaerobic metabolism**.
 - ✓ This can increase to 10% under moderate hypoxia.
- Dysregulation of cardiac metabolism → common diseases that lead to heart failure.



Cardiac metabolism

- The heart requires ATP for:
 - ✓ The function of membrane transport systems (e.g., **Na⁺/K⁺-ATPase**)
 - ✓ Sarcomere **contraction and relaxation**
 - which involves myosin ATPase and
 - ATP-dependent transport of calcium by the sarcoplasmic reticulum.
- Therefore, increasing the mechanical activity of the heart by increasing heart rate and contractility increases myocardial oxidative metabolism

Cardiac energetics



- The energy requirements of the heart are very high, requiring the equivalent of 6 kg ATP/day.
 - ✓ The heart must contract continuously.
- Because the ATP storage within the cardiomyocyte is minimal,
 - ✓ a tight coupling of ATP production and myocardial contraction is essential for normal cardiac function.
 - ✓ It contains negligible energy reserves in the form of glycogen or lipid.

Cardiac energetics...



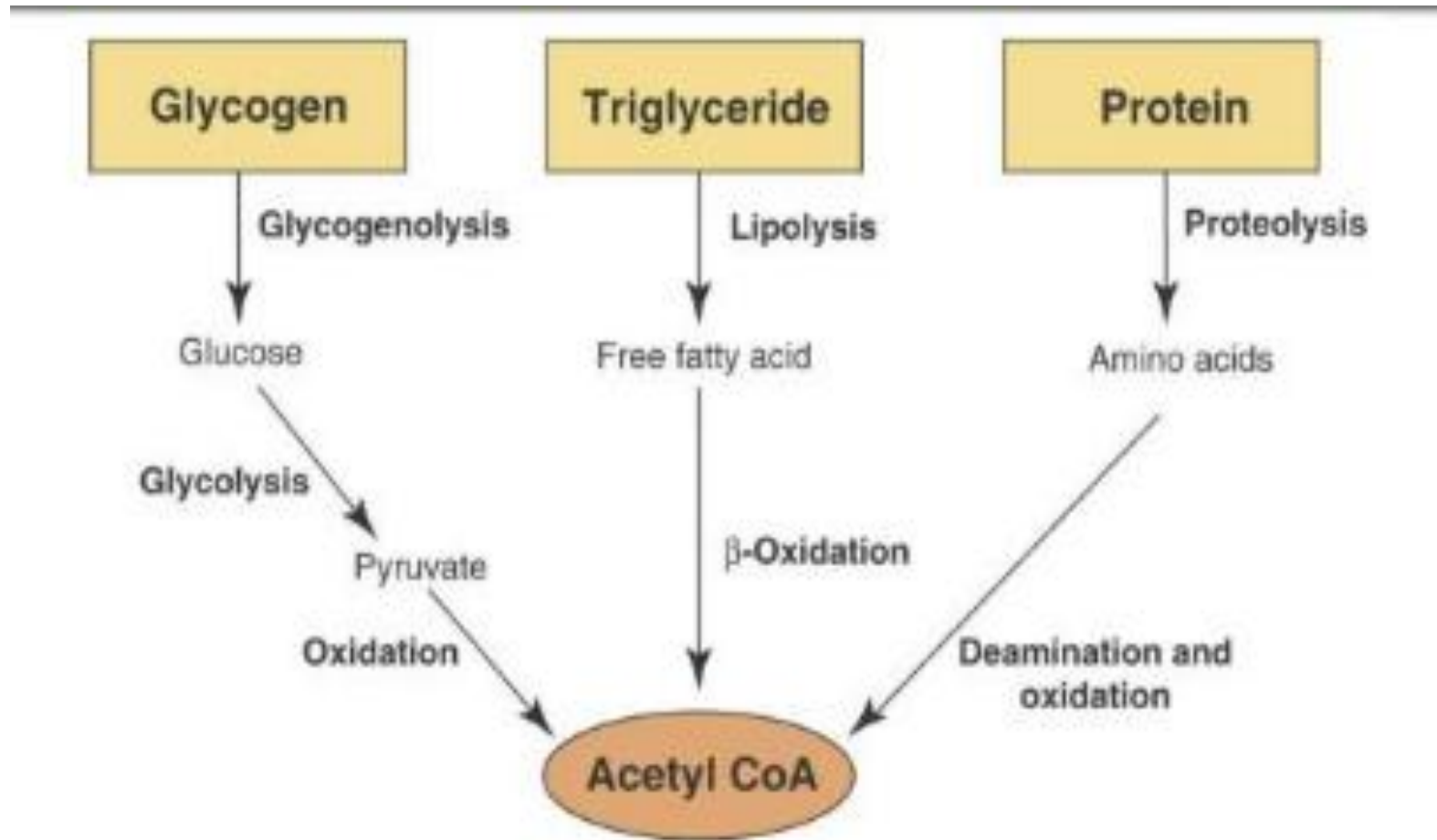
- Cardiac mitochondrion that generates >95% of ATP not only used by the heart, but also **regulates intracellular calcium homeostasis, signaling and cell death.**
- Thus heart muscle, except for short periods of extreme exertion, relies entirely on aerobic metabolism
- Therefore, cardiac myocytes is richly with mitochondria; they comprise up to 40-50% of its cytoplasmic space.
- Thus, the importance of substrate metabolism to cardiac pump function is beyond the scope of only modulation of energy supply.

Substrate metabolism in the Heart



- The heart is capable of using all classes of energy substrates, including **carbohydrates, lipids, amino acids, and ketone bodies**, for ATP production.
- The robustness of cardiac metabolism is reflected by its highest oxygen consumption rate on the per unit weight basis.
- For a human heart, the amount of ATP turned over during a 1-day period is 15 to 20 times of its own weight.

Overall pathway



Preferences

- Sufficient oxygen:
 - Fatty acids (50-70%)
 - Glucose (30%)
 - under ischemic conditions
- Increased muscular activity
 - Glucose and lactate
- Pathological conditions and starvation:
 - Ketone bodies and amino acids

substrate metabolism in the Heart...



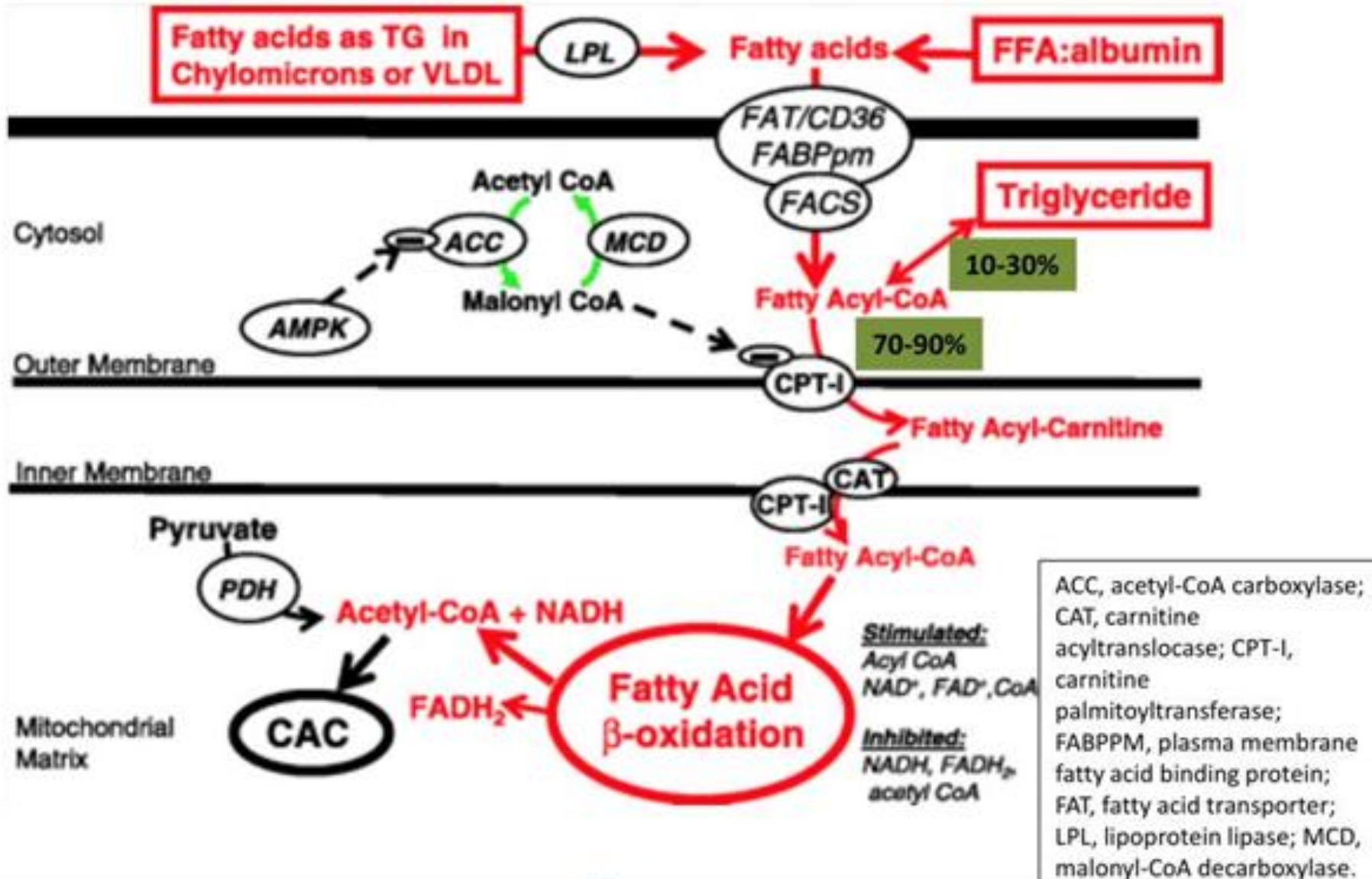
- **Fatty acids** are the resting heart's fuel of choice
 - ✓ but, upon the imposition of heavy work load the heart greatly increases its rate of consumption of the glucose or begin to use glucose,
 - derived mostly from its own relatively limited glycogen store.
- However, it is a metabolic omnivore, changing its substrate utilization according to metabolic demand, oxygen, and substrate availability
 - ✓ crucial for the maintenance of normal cardiac function.

substrate metabolism in the Heart...



- In a normal heart, mitochondria are largely fueled by fatty acyl-coenzyme A & pyruvate,
 - ✓ which are the primary metabolites of FAs and CHO, respectively.
- The entry of long chain acyl-CoA into the mitochondrion is with the rate-limiting step at the **carnitine-palmitoyl transferase-1 (CPT1)** reaction.
- The oxidation of pyruvate is regulated at the pyruvate dehydrogenase (PDH) reaction.

Fatty acid metabolism



Fatty Acid Oxidation



- Several mechanisms are involved in the uptake of long-chain fatty acids into the myocytes, including the breakdown of triglycerides by lipoprotein lipase, **fatty acid translocase (CD36), membrane fatty acid-binding protein, and fatty acid transport protein 3.**
- Within the cardiac myocytes, long-chain fatty acids are converted to long-chain fatty acid acyl CoA by the enzyme fatty acid **acyl CoA synthase**, but unlike short chain fatty acids require a specific transport system (the carnitine shuttle) to enter the mitochondria.

Fatty Acid Oxidation.....

- On the outer mitochondrial membrane, the enzyme **carnitine palmitoyltransferase type 1 (CPT-1)** catalyzes the addition of a carnitine group to the long-chain fatty acid acyl CoA which crosses to the opposite side of the mitochondrial membrane in exchange for CoA, where the enzyme CPT-2 cleaves the carnitine off.
- CPT-1 is the rate limiting step in long-chain fatty acid metabolism and is potently inhibited by **malonyl CoA produced from acetyl CoA by the actions of the biotin-dependent enzyme acetyl CoA carboxylase** located in the endoplasmic reticulum.
- Another enzyme, malonyl CoA decarboxylase (MCD) catalyzes the reverse reaction.

Fatty Acid Oxidation.....



- In the mitochondrial matrix, the long-chain acyl CoA fatty acids undergo β -oxidation, comprising a recurring series of 4 reactions culminating in the release of acetyl CoA.
- The β -Oxidation Pathway: **Dehydrogenation, Hydration, Dehydrogenation, & Thiolase Reaction/Cleavage**
- During each of these cycles:
 - ✓ FAD is converted to FADH₂ and
 - ✓ NAD into NADH, producing reducing equivalents for the electron transport chain.
 - ✓ Each pass removes one acetyl moiety in the form of acetyl-CoA
 - Acetyl CoA enters the TCA cycle.

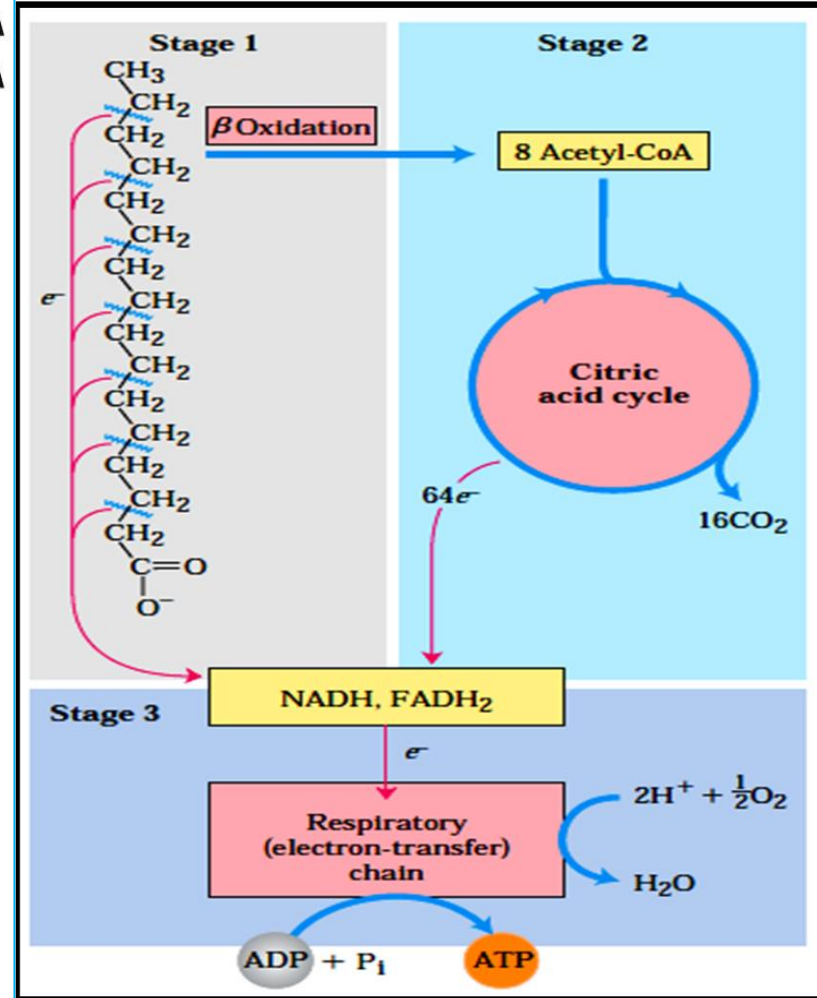
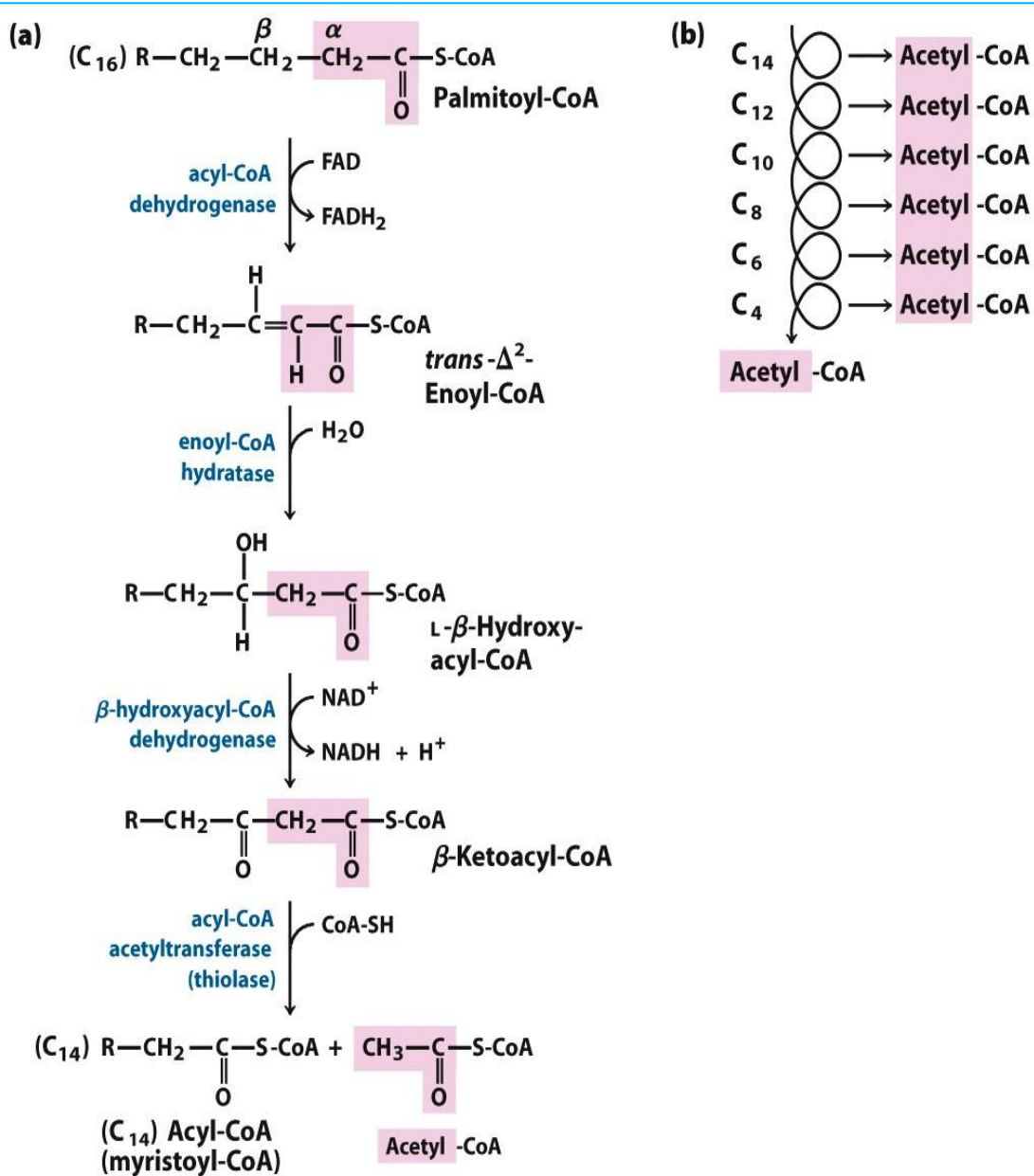
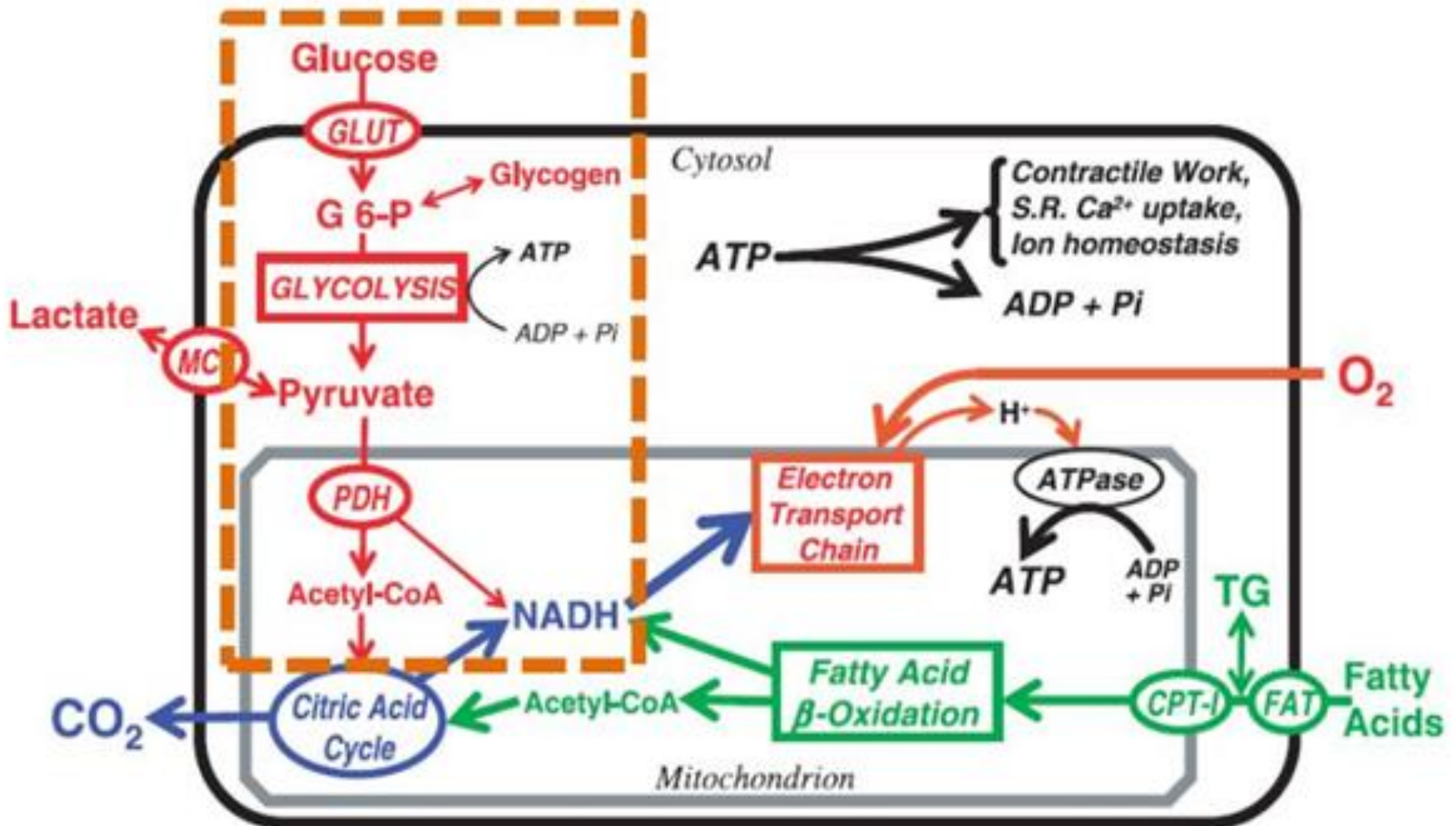


Figure 17-8
 Lehninger Principles of Biochemistry, Fifth Edition
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Glucose metabolism



Glycolysis and Carbohydrate oxidation



- After uptake into the cytosol glucose is converted via a series of 10 intermediate steps in the cytosol (glycolysis) to pyruvate which is actively transported into the mitochondria .
- The pyruvate dehydrogenase (PDH) enzyme complex in the mitochondrial matrix catalyzes the conversion of pyruvate to acetyl coenzyme A (CoA).
- PDH is a rate-limiting step in carbohydrate oxidation, being negatively regulated by kinase and positively regulated by phosphatase enzymes and subject to both
 - ✓ Positive(upstream, by pyruvate) and
 - ✓ Negative (downstream, by acetyl CoA/CoA, ATP/ADP, & NADH/NAD⁺ ratios) allosteric regulation.

Glycolysis and Carbohydrate oxidation...



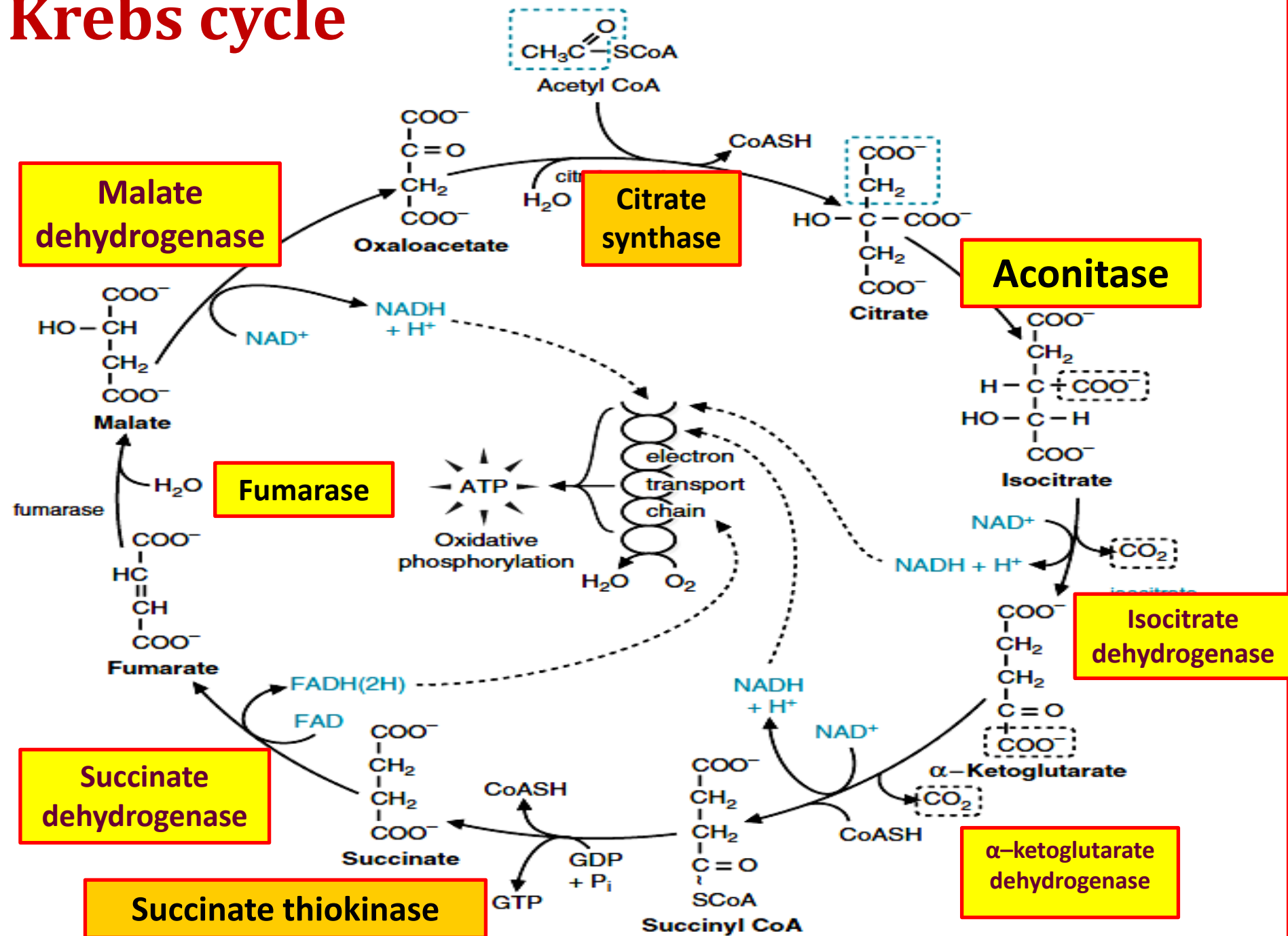
- Hypoxia inhibits **PDH via hypoxia-inducible factor stabilization**, therefore pyruvate is converted to lactate by the enzyme lactate dehydrogenase, but this only generates 2 molecules (net) of ATP per molecule of glucose metabolized.
- Hypoxia also has more generalized effects on cardiac metabolism such as including
 - ✓ modulating the enzyme activities of Krebs cycle, and
 - ✓ controlling the mitochondrial permeability transition pore opening.

Glycolysis and Carbohydrate oxidation...



- Under aerobic conditions, acetyl CoA generated from pyruvate enters the TCA cycle in the mitochondrial matrix.
- One unit of acetyl CoA generate 3 NADH from NAD⁺, which acts as an electron donor to complex I of the electron transport chain (see below).
- The oxidation of succinate to fumarate by succinate dehydrogenase (which is also complex II of the electron transport chain) also donates electrons to the electron transport chain .

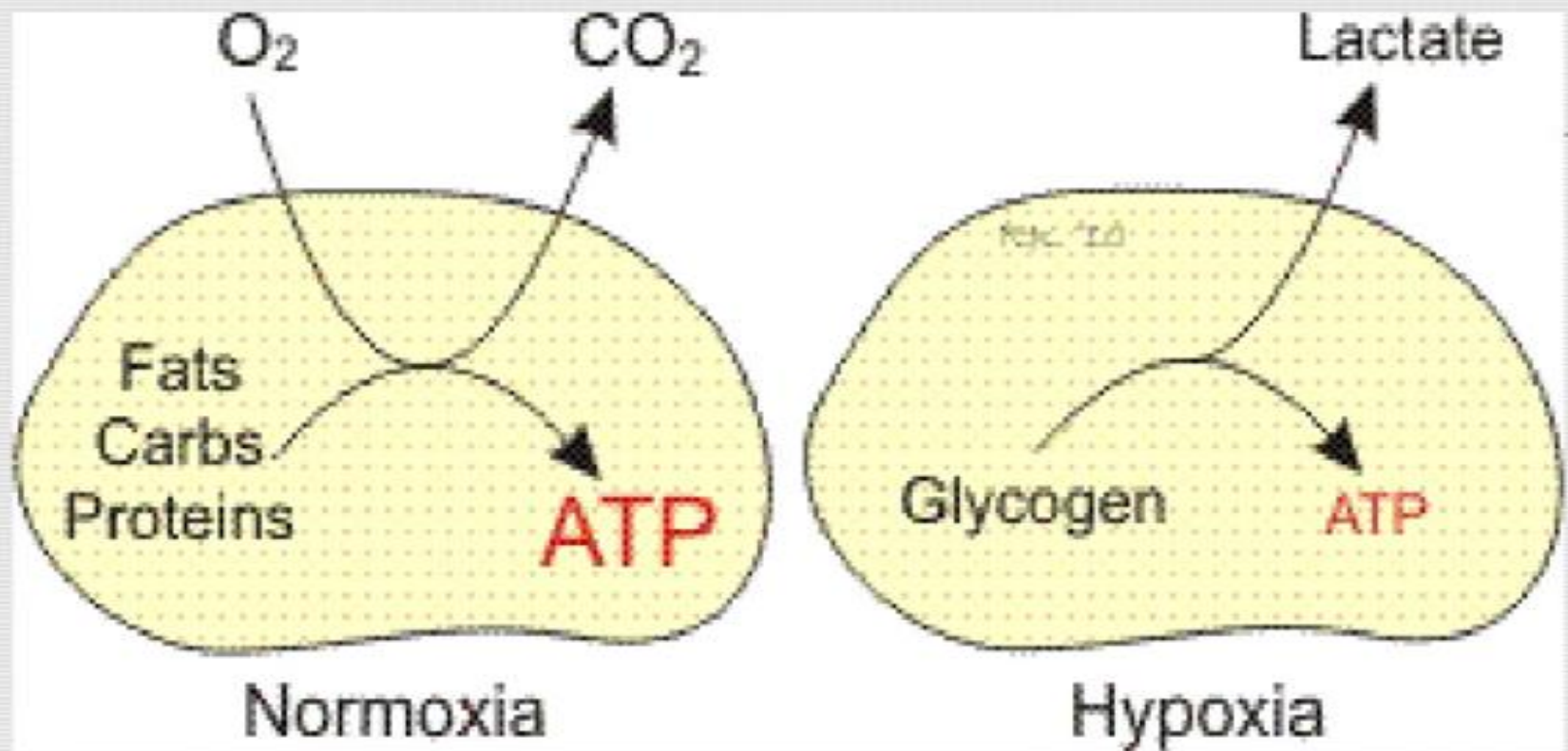
Krebs cycle



Substrate metabolism in the Heart...



- Other substrates, including **lactate, ketone bodies, and amino acids**, can enter mitochondria directly for oxidation.
- Metabolism of ketone bodies yields acetyl-CoA, whereas amino acid catabolism yields keto-acids, which are further metabolized to enter the tricarboxylic acid cycle.
- The contribution of ketone bodies and amino acids to overall cardiac oxidative metabolism is considered to be minor because of the low availability of these substrates under normal physiological conditions.



Substrate metabolism in the Heart...

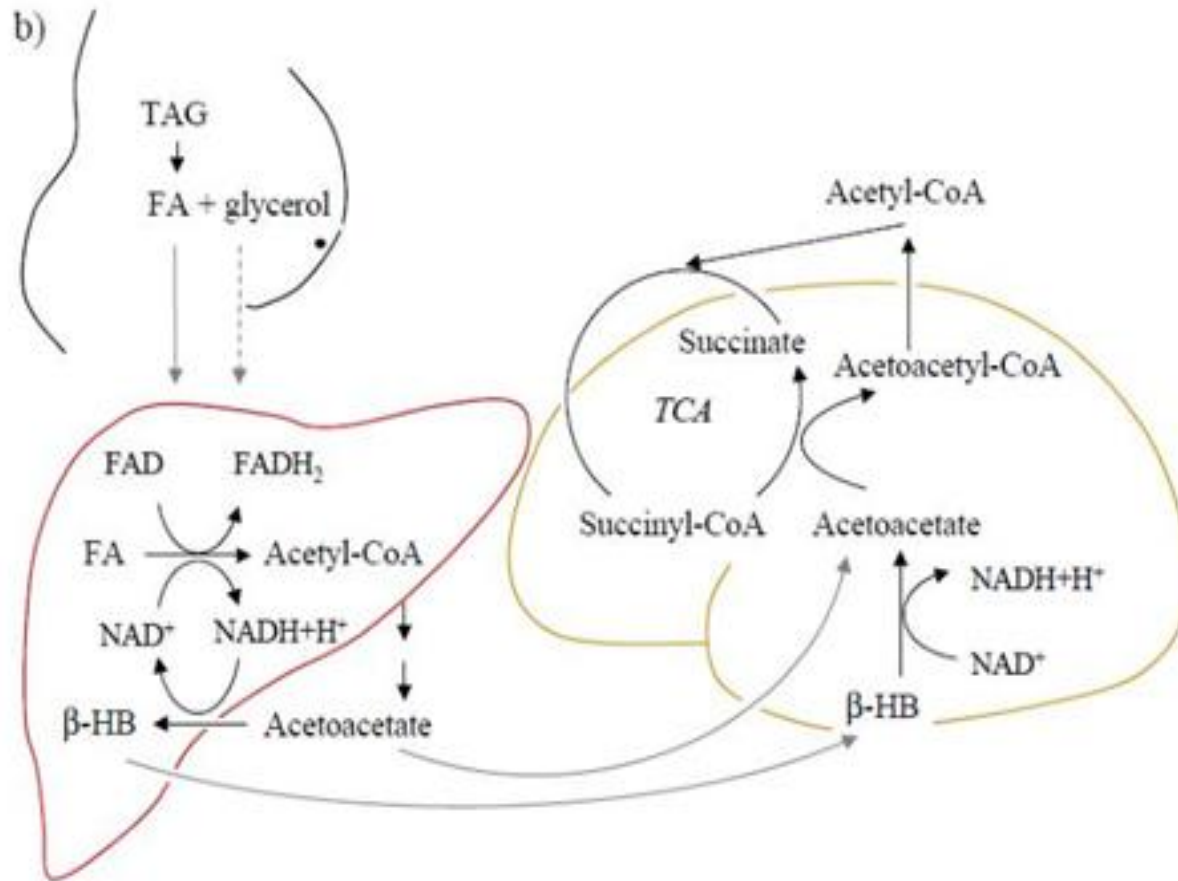


- Prolonged fasting or ketogenic diet increases the blood level of ketone bodies and results in enhanced use by the heart.
- On the other hand, cardiac extraction and oxidation of lactate becomes predominant during exercise as skeletal muscle lactate production increases.
- In isolated perfused hearts, the addition of lactate or ketone bodies in the perfusate significantly reduces the oxidation of glucose and fatty acids.

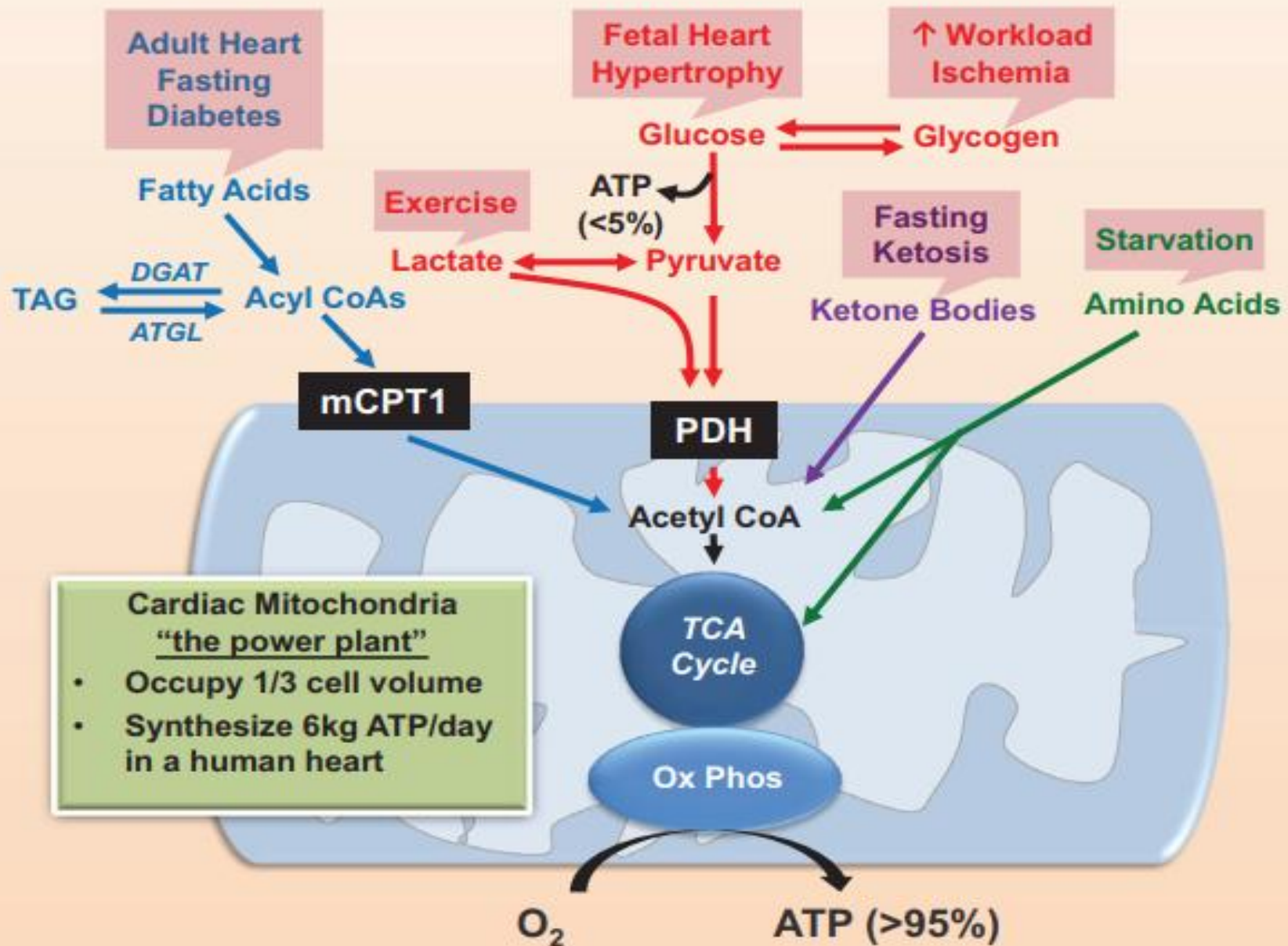
Starvation



Production of ketone bodies

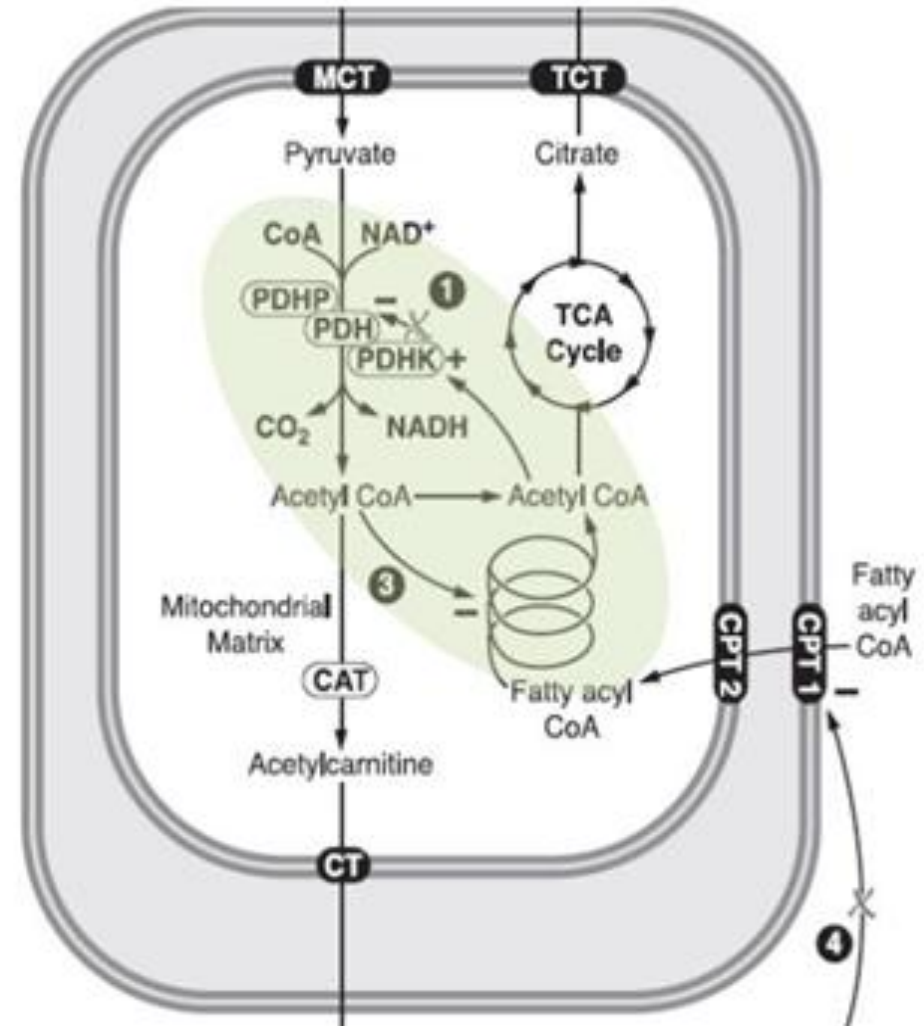


Overview of the metabolic network



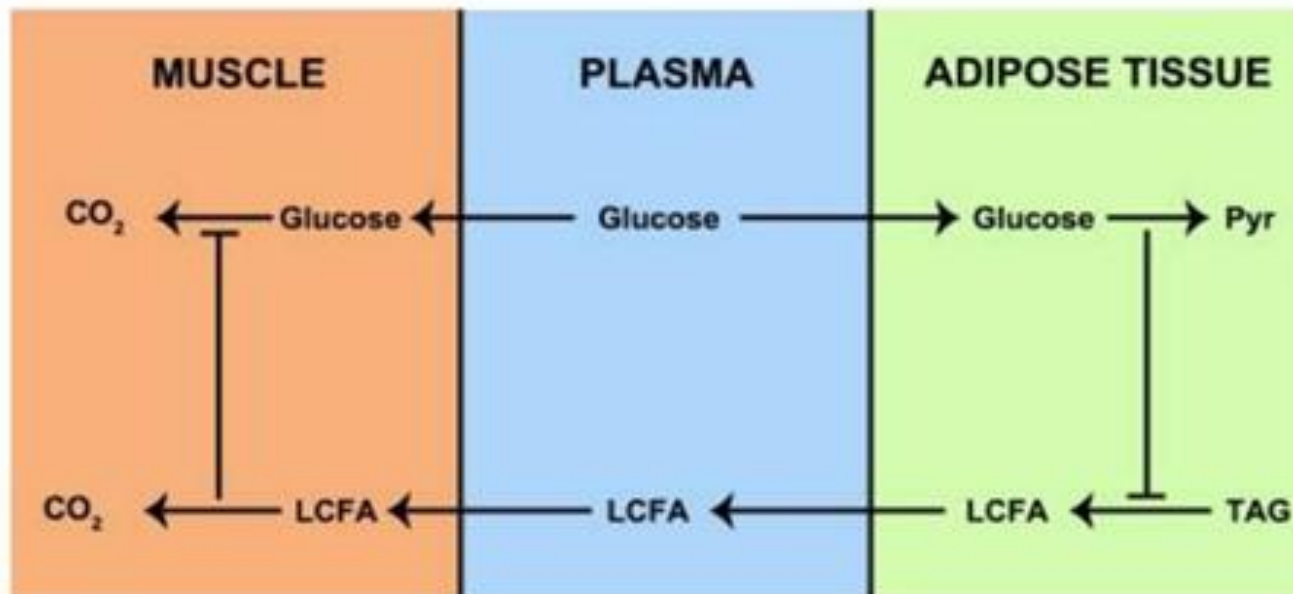
Glucose-fatty acid (Randle) cycle

- The Randle cycle describes the reciprocal relationship between fatty acid and glucose metabolism.
- The increased generation of acetyl CoA derived from fatty acid-oxidation decreases glucose (pyruvate) oxidation.
- The increased generation of acetyl CoA derived from glucose (pyruvate) oxidation inhibits fatty acid -oxidation



Glucose-fatty acid (Randle) cycle.....

- The “cycle” also describes the control of fuel selection through the dynamic interactions between circulating concentrations of glucose and fatty acids in coordination with hormones.
- *Inhibition of glucose utilization by fatty acids is a form of glucose intolerance that resembles, or may lead to, insulin resistance.*

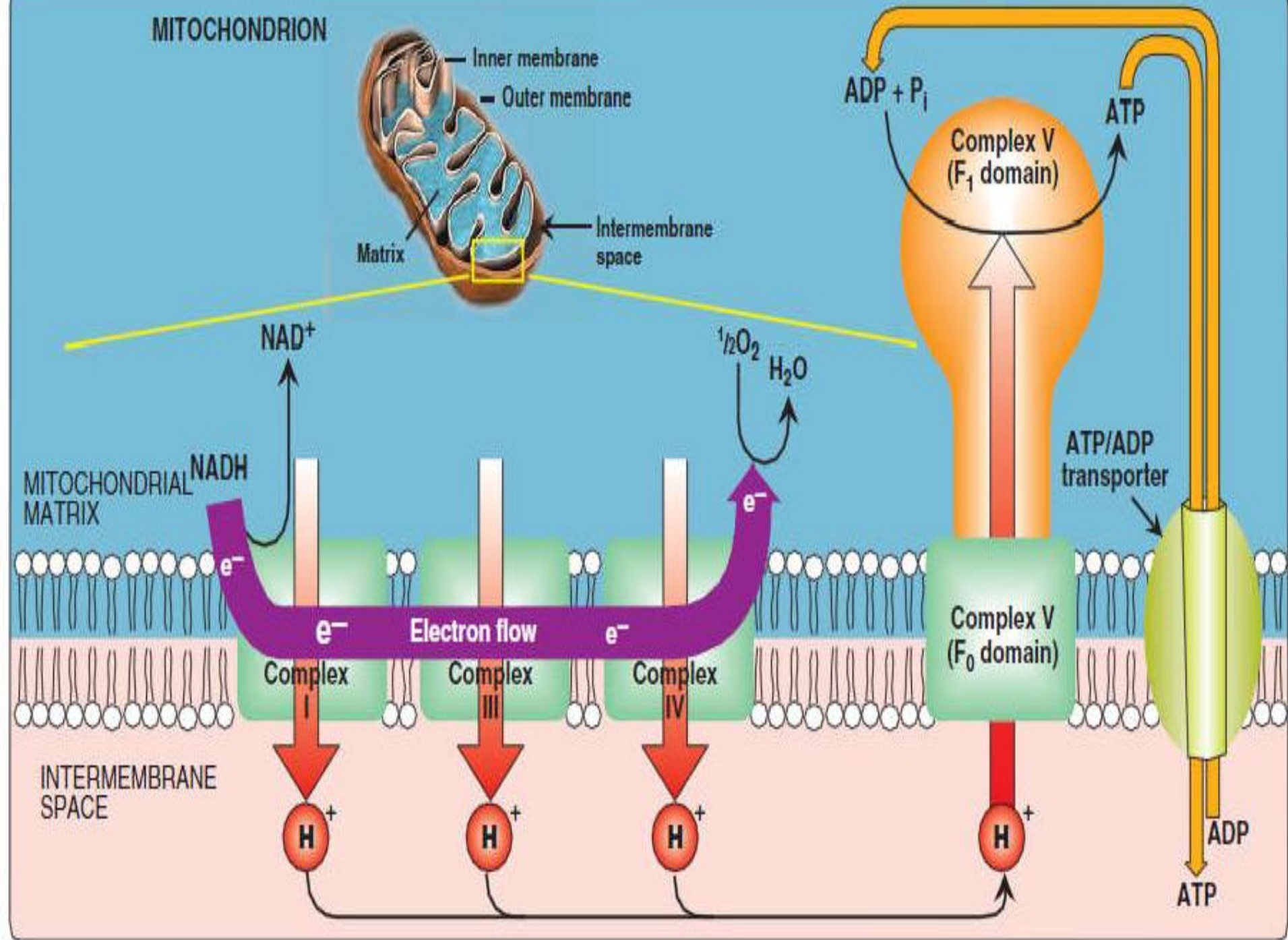
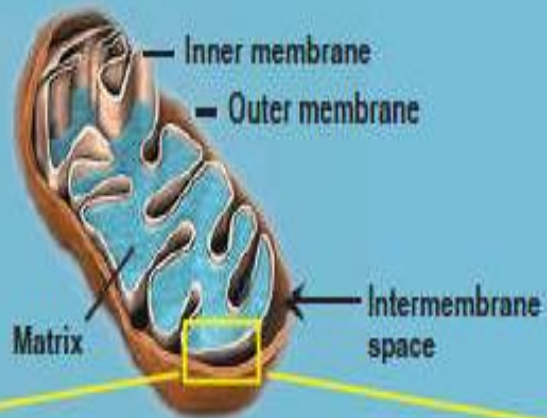


Generation of ATP via the ETC



- At the inner mitochondrial membrane, a series of redox reactions occur in which electrons are donated to and then transferred from one complex to another.
- The energy derived by these electron transfers is used to pump protons from the matrix into the Intermembrane space, generating an electrochemical gradient across the inner mitochondrial membrane.
- Complex V contains an ion channel component (F_o) through which protons flow back into the mitochondrial matrix, releasing energy that drives the phosphorylation of ADP to ATP by the F_1 component of the complex.

MITOCHONDRION



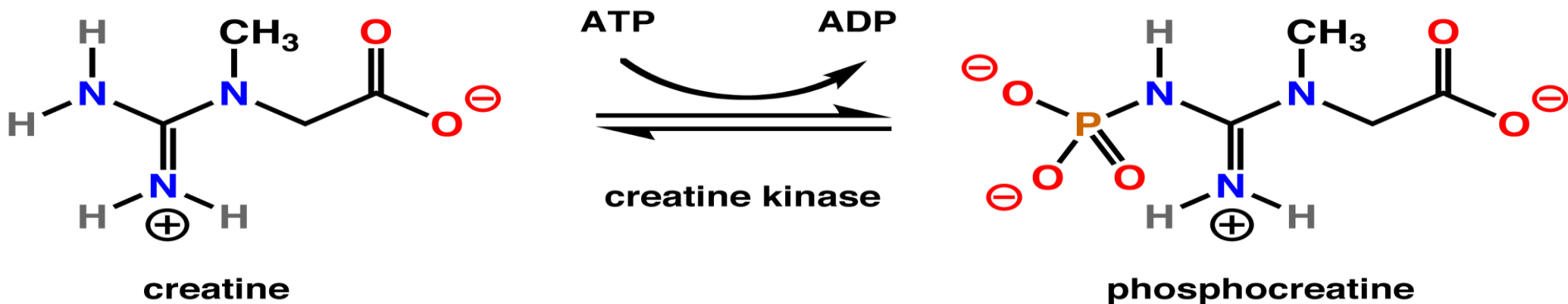
Generation of ATP via the ETC ...



- In health, minor electron leakage ($<1\%$) may occur during these electron transfers (particularly at complexes I and III), resulting in the generation of superoxide but in disease states may be greater, contributing substantially to increased oxidative stress.
- Theoretically, 38 molecules of ATP can be generated by the complete oxidation of 1 molecule of glucose through the above processes, however, in practice, the net yield is somewhat less, in part because of some degree of uncoupling of the electron transport chain.
- In the heart muscle, uncoupling protein 3 (and to a lesser extent, uncoupling protein 2) is expressed, and this is increased when plasma free fatty acids are elevated.

Energy Transfer

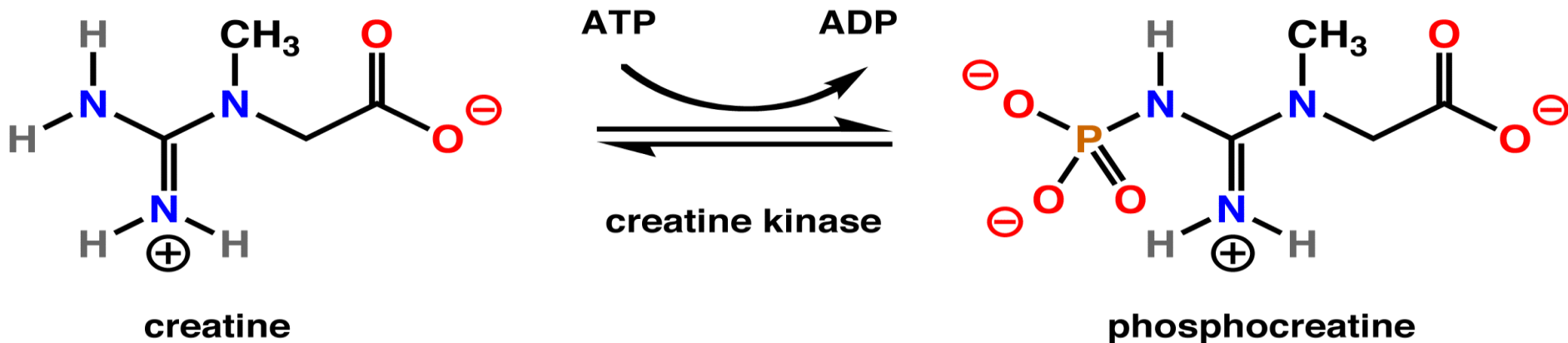
- ATP is generated in the mitochondria but is mainly consumed elsewhere principally by sarcomeric proteins and ion channels.
- The predominant mechanism of energy transfer is the **creatine kinase (CK)** system.
- Mitochondrial CK generates **phosphocreatine** via the phosphorylation of creatine.
- Phosphocreatine is transported to cellular sites of energy consumption, where a reversal of this reaction by the cytosolic isoform of CK releases ATP.



Energy Transfer...



- The forward reaction occurs within the mitochondrial intermembrane space.
- The resultant **PCr** is relatively less polar and can store to high levels in the cytosol.
- **PCr** available for rapid renewal of ATP at times of high need.
- This reverse reaction is catalyzed by cytosolic CKs



Regulation



- Regulation of fatty acid and carbohydrate metabolism is described by the phenomenon known as **Randle Cycle**
 - ✓ whereby increased fatty acid utilization results in **increased acetyl CoA** production, thereby **inhibiting PDH activity**, and increases **cytoplasmic citrate**, which in turn **inhibits glycolysis**.
 - ✓ Conversely, when carbohydrate metabolism is **increased**, **citrate** from the TCA cycle is exported to the cytoplasm where it is converted to acetyl CoA and then to malonyl CoA with consequent **inhibition of CPT-1** activity, thereby reducing fatty acid metabolism.

Regulation cont...

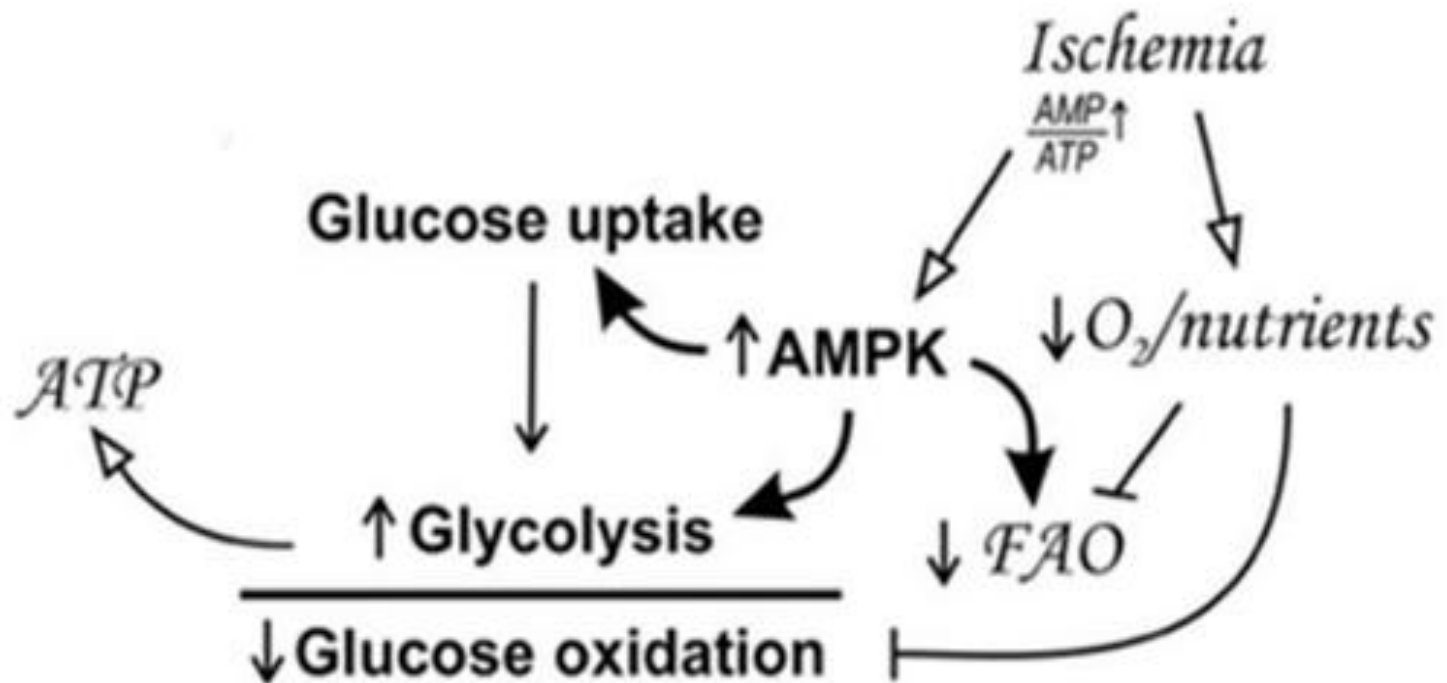


➤ Energy Sensing

- ✓ An exquisite system of energy sensing exists in the heart to match energy production to energy demand.
- ✓ It central to this is the AMP kinase enzyme.

How does ischemia alter metabolic profile?

- Ischemia results in
 - Decrease of O_2 and nutrients, which inhibit fatty acid oxidation
 - Increase in AMP/ATP ratio, which activates AMPK



Atherosclerosis & Dyslipidemias

Outline

- ✓ Mechanism of atherosclerosis
- ✓ Dyslipidaemia and their type



I. Atherosclerosis

- ✓ **Lipoproteins** distribute **triacylglycerols** and **cholesterol** between the **intestine, liver, and peripheral tissues**
- ✓ **Abnormal lipoprotein metabolism** is the principal factor in the development of **atherosclerosis**
 - ❖ a process that underpins **arteriosclerotic cardiovascular disease** (ASCVD), including:-
 - coronary heart disease,
 - stroke, and
 - peripheral vascular disease

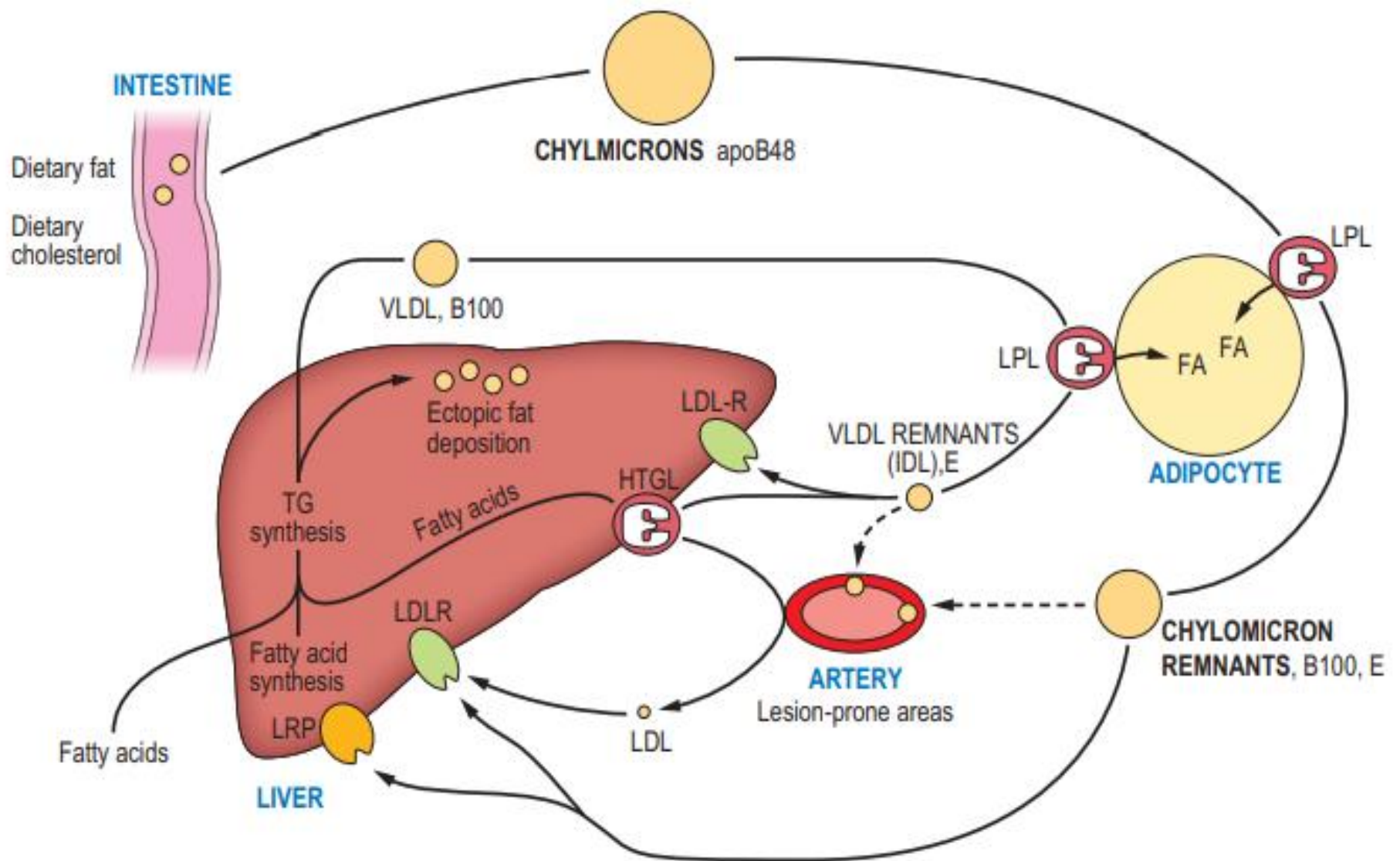
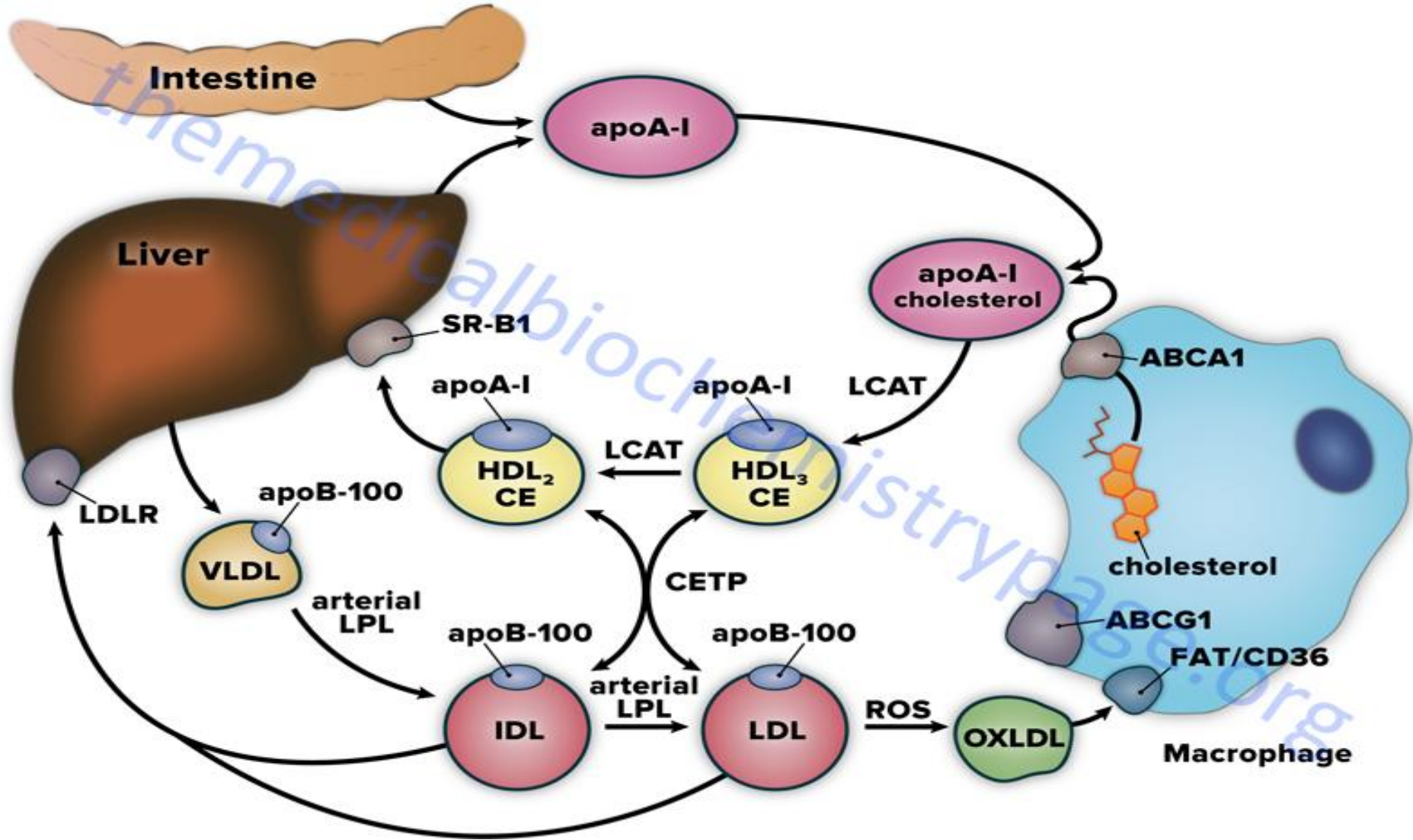
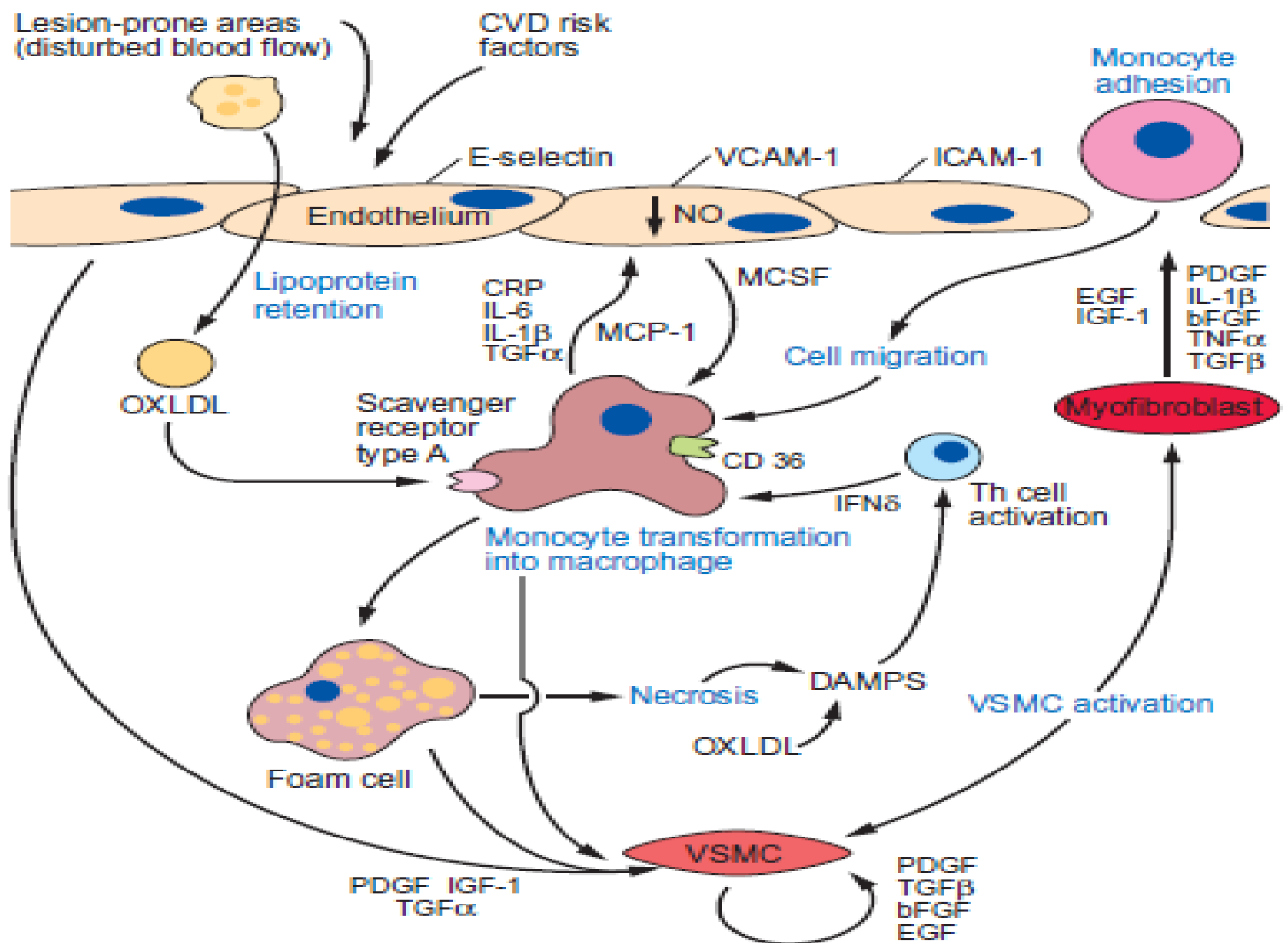


Figure: Lipoprotein metabolism: the fuel distribution stage.

Atherosclerosis cont...

- ❖ **Atherosclerosis**: is a pathological condition in which deposition of lipid-rich structures (atheromatous plaques) deposit in the walls of arteria.
- ❖ Surface fixation of lipid-rich layers facilitates deposition of additional layers; a potential clogging site slowly forms.
- ❖ In the myocardium, this condition may cause a heart attack (myocardial infarction); in the brain, it may cause stroke. Both are very serious life-threatening conditions
- ❖ **Atherogenesis** is a process that leads to the narrowing, or a sudden complete occlusion, of the arterial lumen





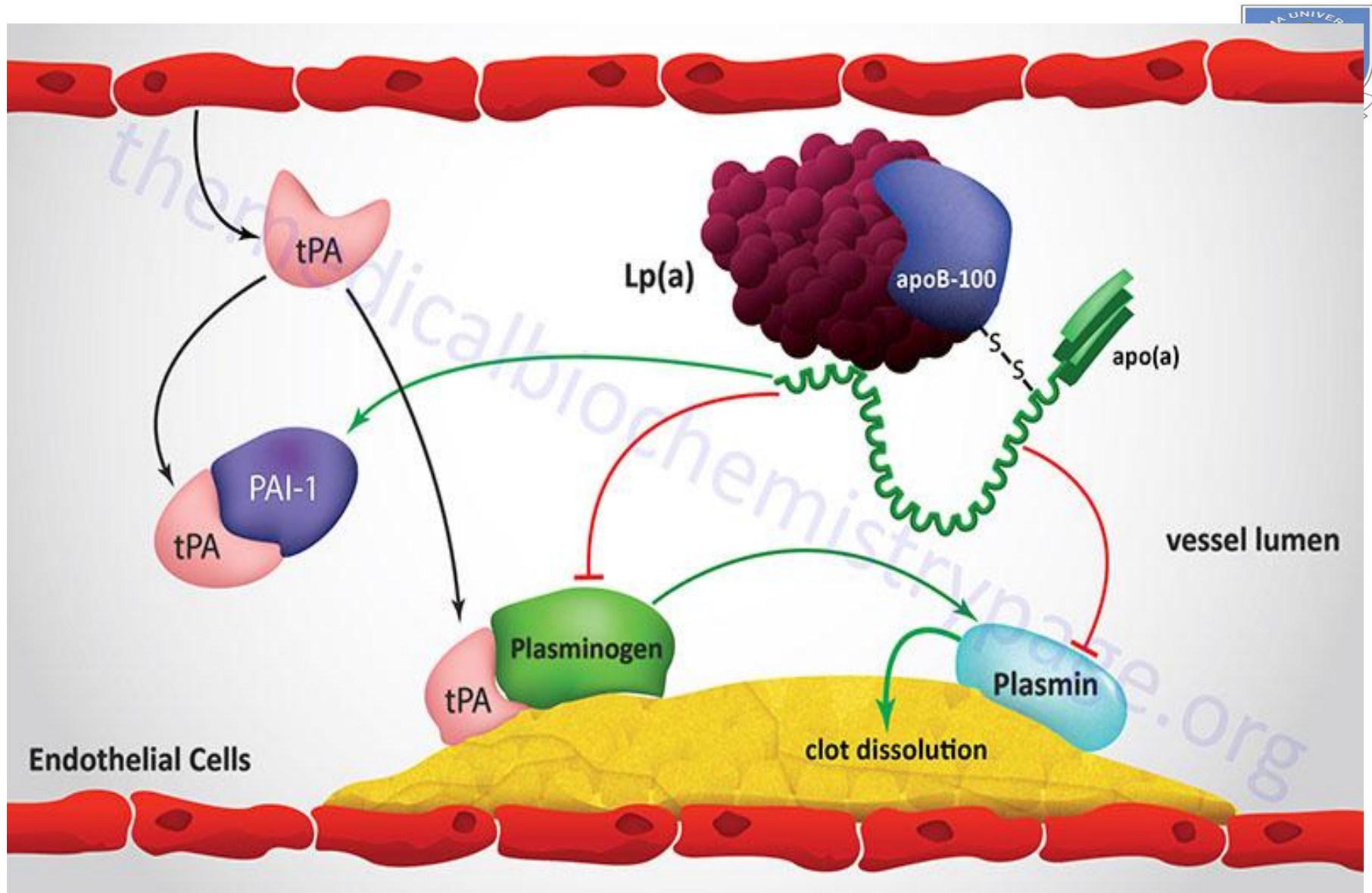


Figure: Role of Lp(a) in promotion of atherogenesis.

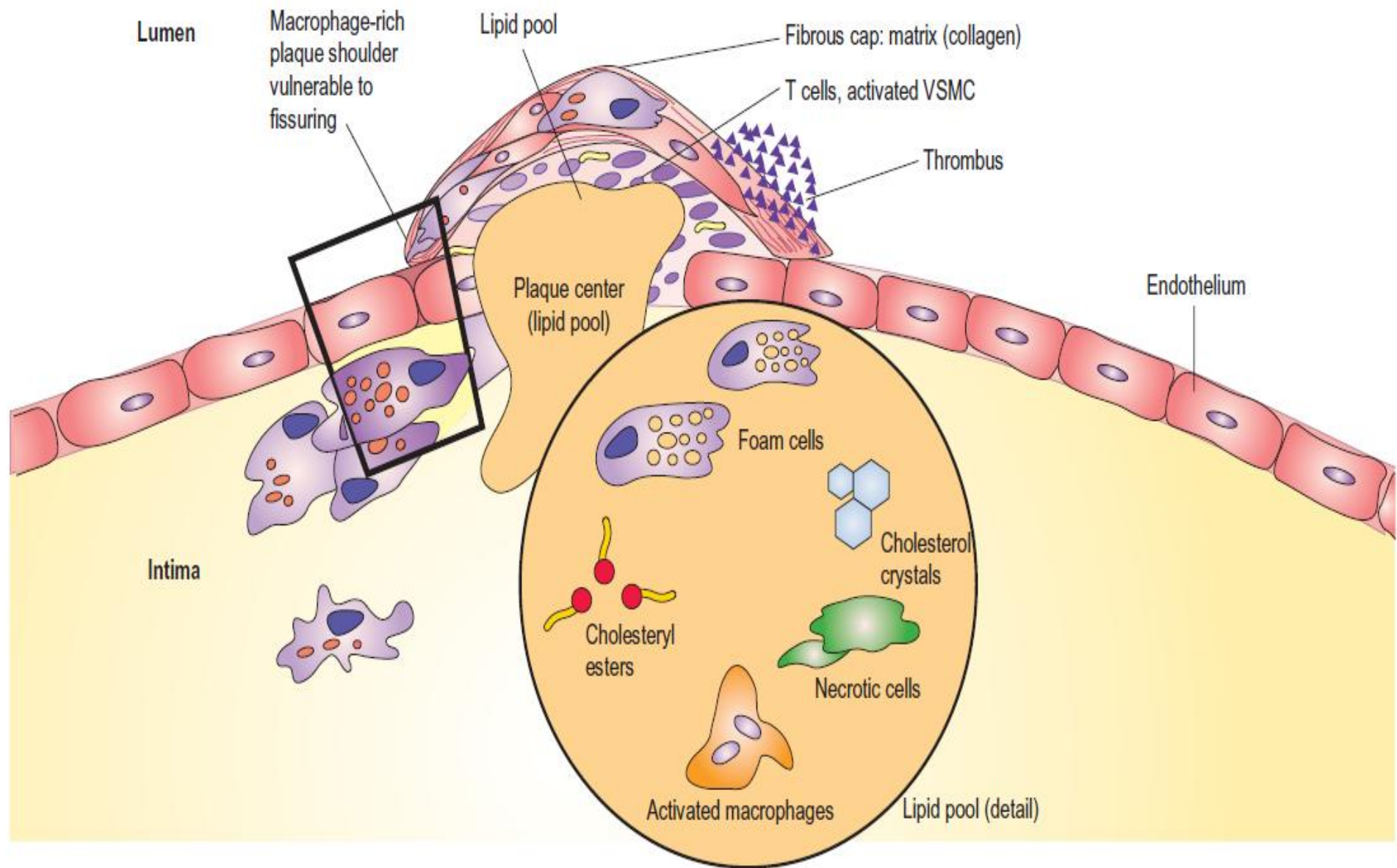
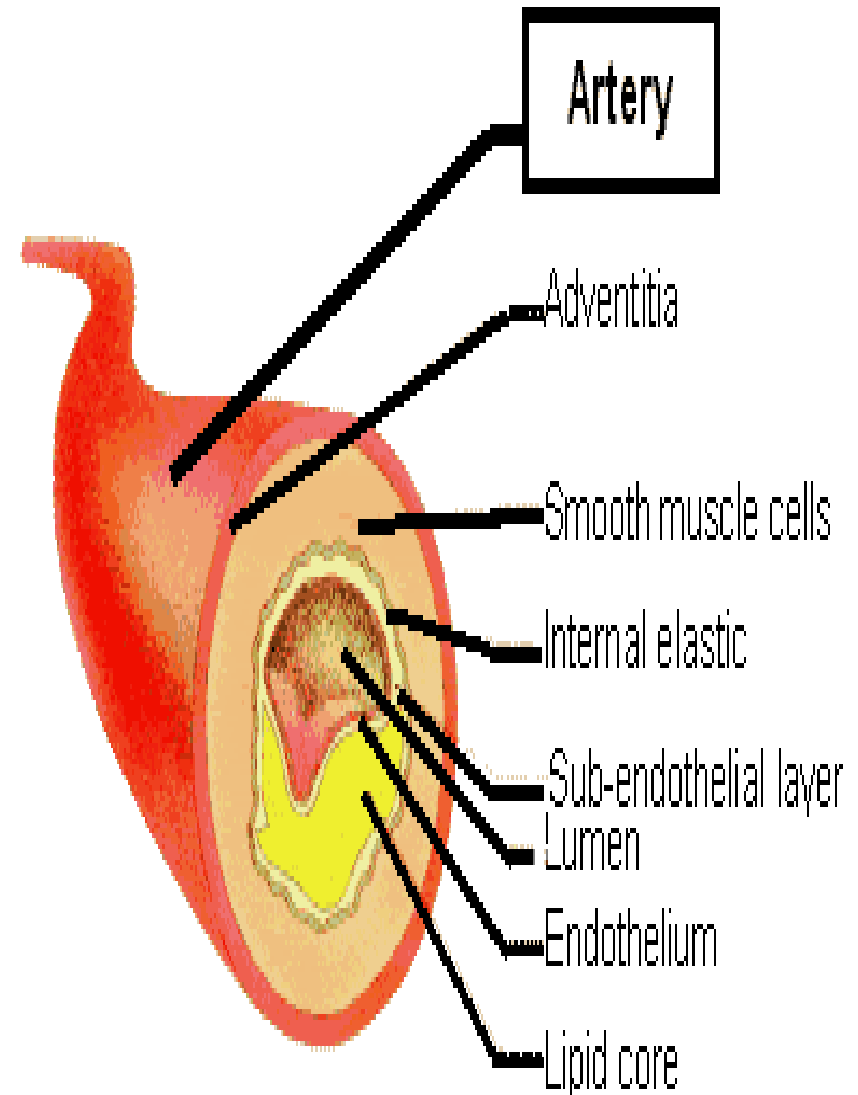
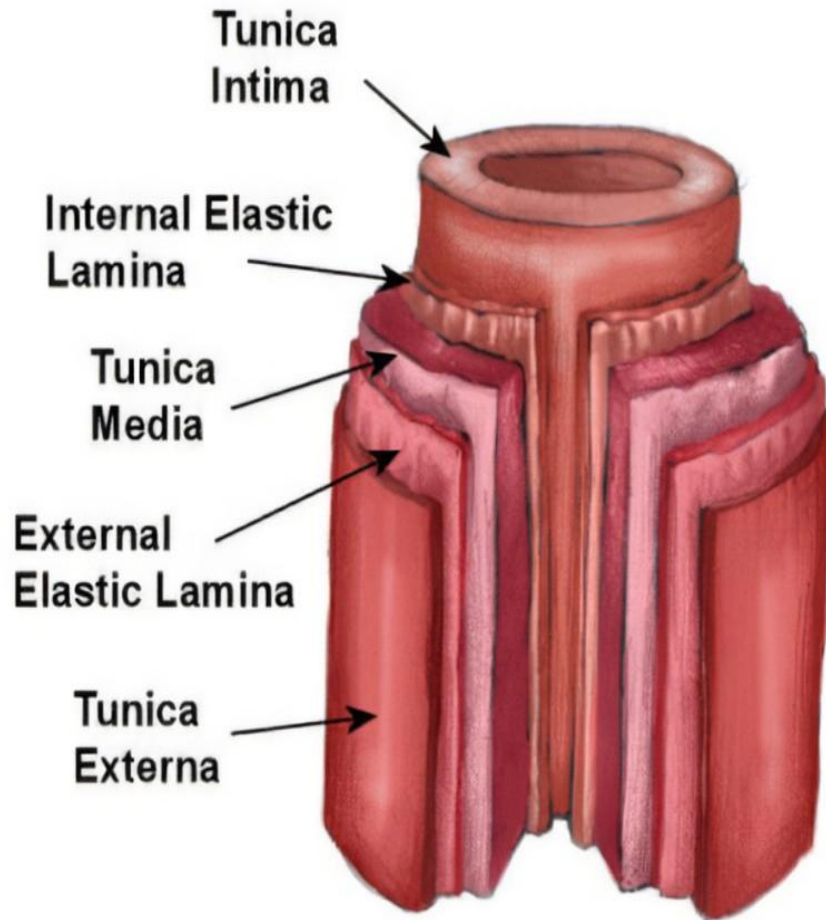


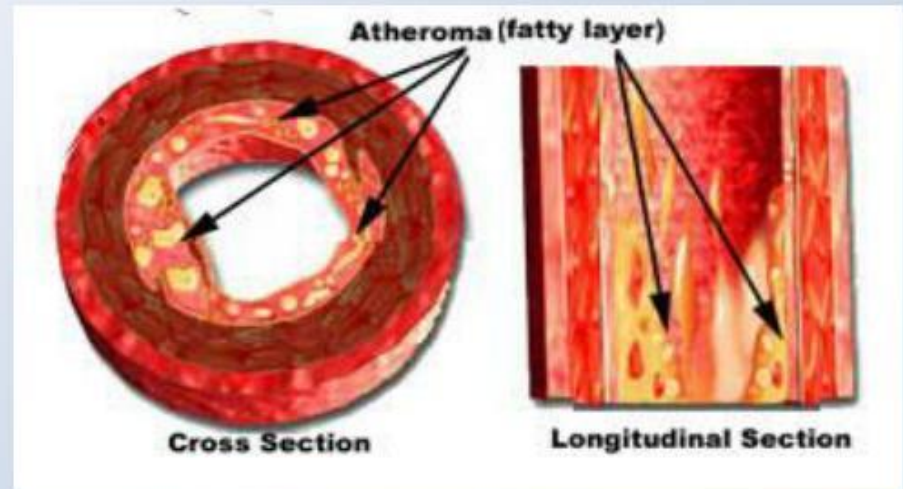
Figure: Atherosclerotic plaque. Mature atherosclerotic plaque

Atherosclerosis

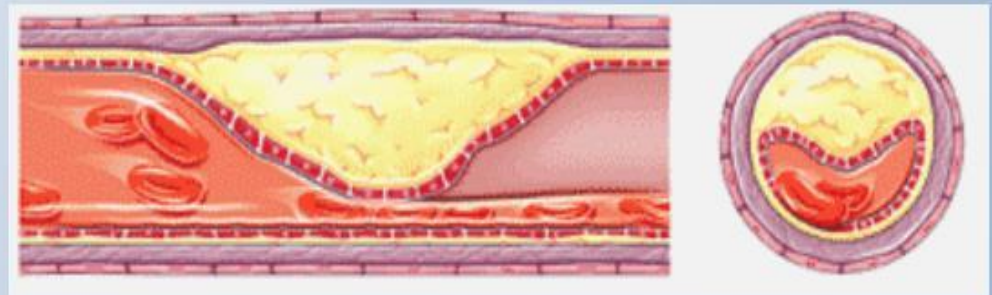


Atheroma

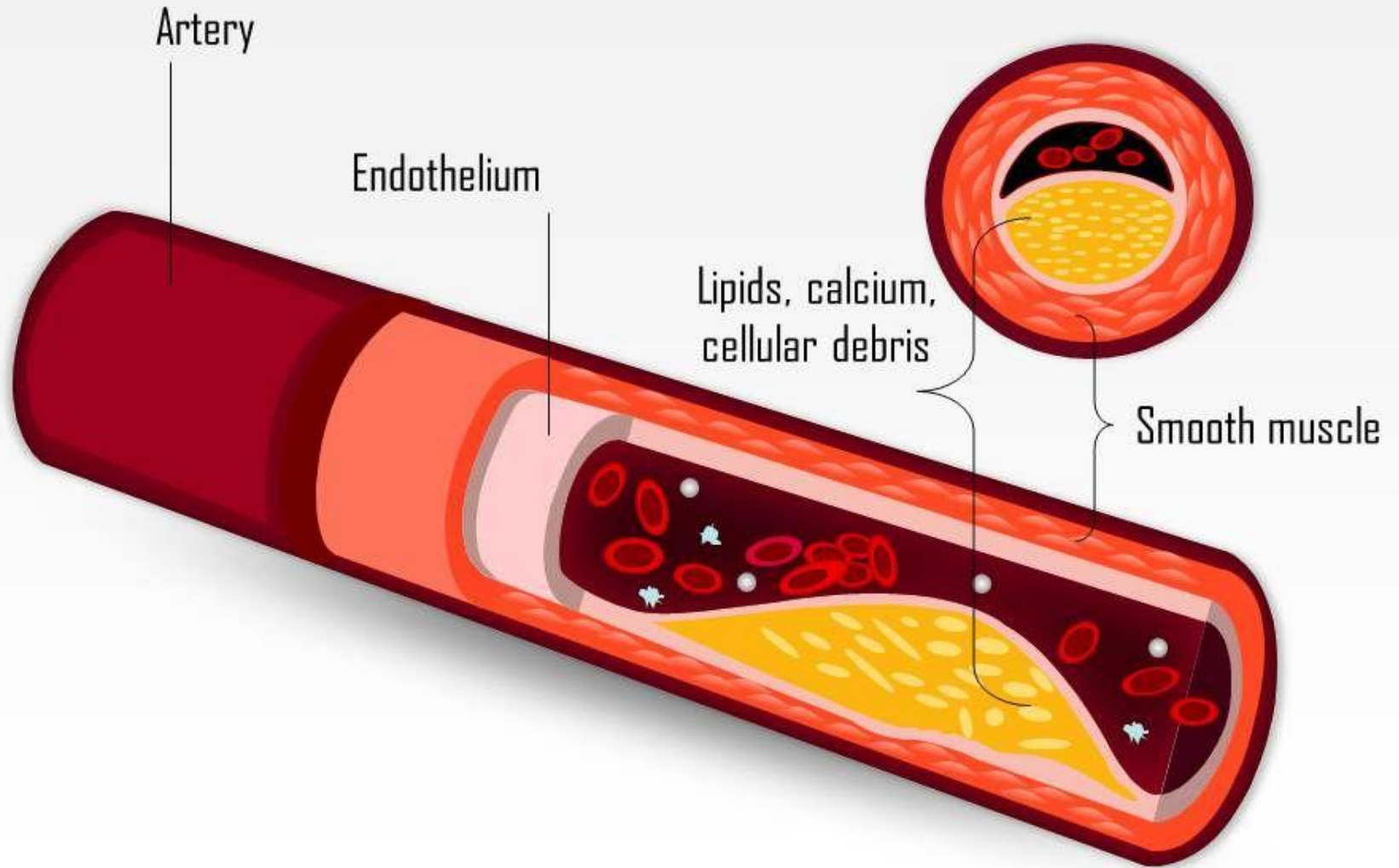
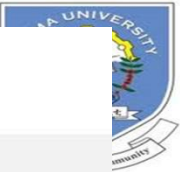
- **Fatty deposits** build up under the endothelium of the artery when it becomes damaged.
(damage can be caused by uneven blood flow, high blood pressure, chemicals or viral infection.)



- White blood cells collect under the endothelium and absorb fatty materials, e.g. LDLs (contain **cholesterol**).



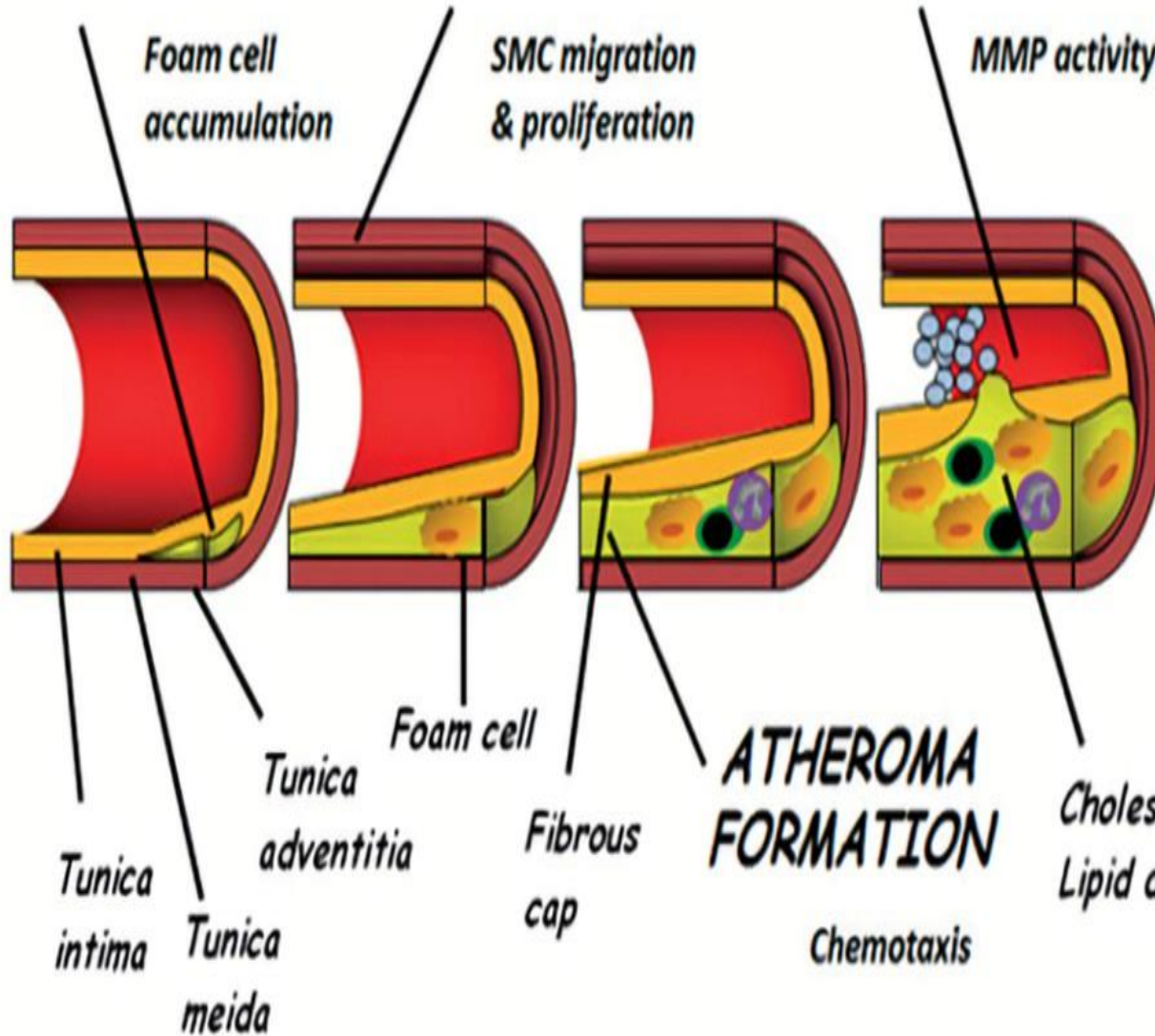
Atherosclerosis



LIPID RETENTION

INTIMAL HYPERPLASIA

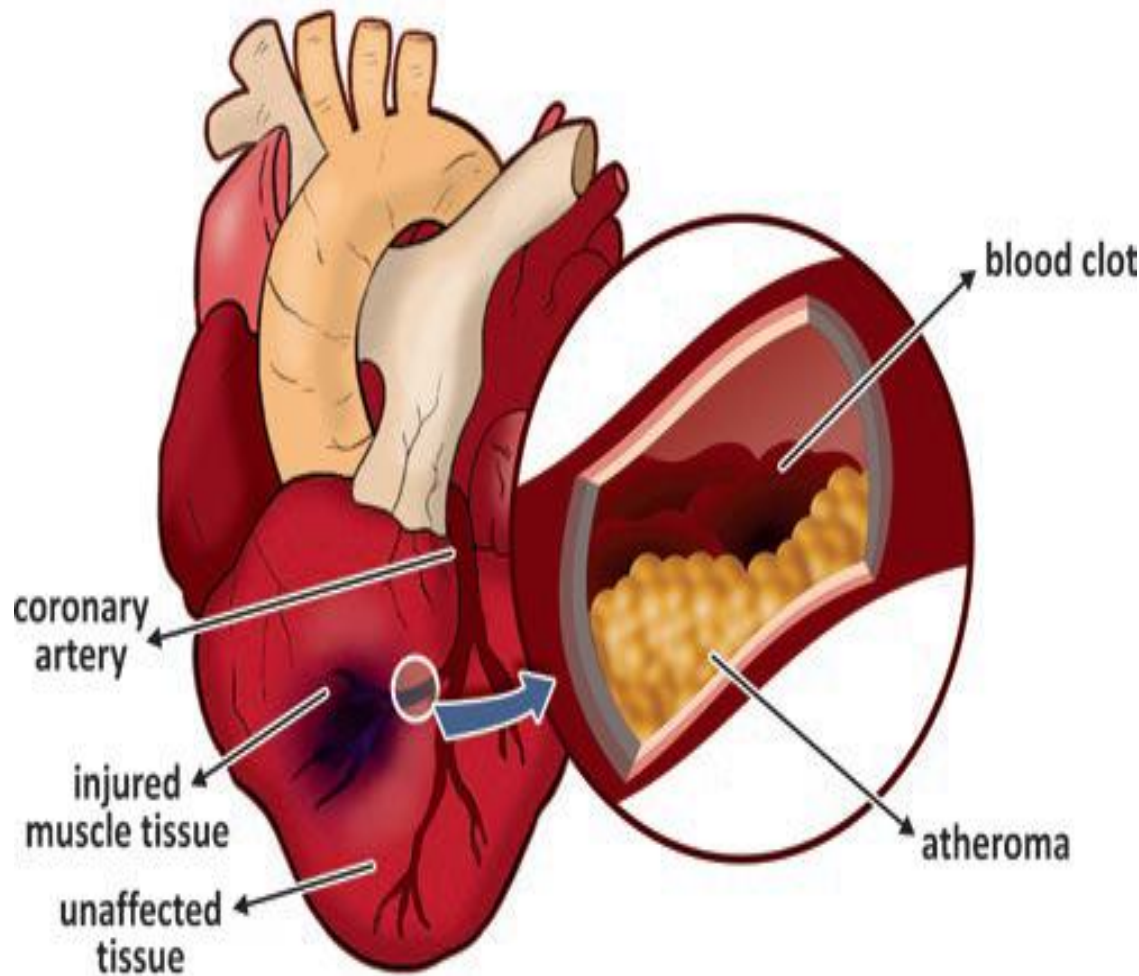
PLAQUE RUPTURE



CEREBRAL & MYOCARDIAL ISCHEMIA

I/R Injury





- ❖ Clogging of a heart artery with lipid-rich materials
- ✓ may cause total interruption of blood flow and consequent supply of oxygen and nutrients to tissues.
- ❖ Cells die forming a lesion in the heart's muscle.

Biomarkers of Cardiac Injury



What are biomarkers?

- A biomarker is a substance used as an indicator of the biologic state.
- Characteristic objectively measured and evaluated as an indicator of
 - ✓ normal biologic processes,
 - ✓ pathogenic processes, or
 - ✓ pharmacologic responses to a therapeutic intervention.



What are Cardiac biomarkers?

- Are protein molecules released into the blood stream from damaged heart muscle
 - ✓ with characteristic rise and fall pattern
- **Cardiac marker**: a clinical laboratory test useful in cardiac disease, most commonly for detecting myocardial injury
- **Characteristics of ideal cardiac marker**
 - ✓ High cardiac specificity
 - ✓ Easy diagnosis
 - ✓ Marker should play a designed role in the treatment and management of clinical subject.



Classification of cardiac biomarkers

1. Biomarkers of myocardial injury

- ✓ Creatine kinase-MB
- ✓ Myoglobin
- ✓ Cardiac troponins

2. Biomarkers of hemodynamic stress

- ✓ Natriuretic peptides

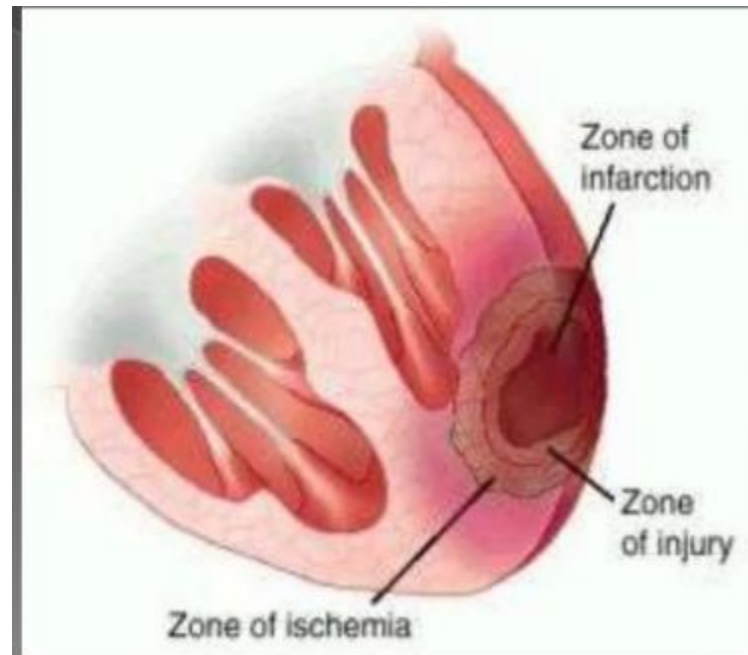
3. Inflammatory and prognostic biomarkers

- ✓ C-reactive protein
- ✓ Myeloperoxidase
- ✓ Homocysteine

Biomarkers of myocardial injury

Myocardial infarction is due to:

- Occlusion of coronary arteries → restricted blood supply (oxygen) to heart tissue (**ischemia**) → damage to heart tissue (**infarction**) → release of enzymes and other proteins into the blood.



Creatine kinase



- Is an enzyme expressed in a number of tissues
- It catalyzes the conversion of creatine to phosphocreatine.
- Consists of two subunits: B (brain type) or M (muscle type).
 - Making three different isozymes

Multiple Forms of Creatine Kinase (CK):

Form	Description
CK-MM	predominates in the skeletal muscles.
CK-MB	predominates in the cardiac muscle .
CK-BB	predominates in the brain, smooth muscles, and lungs.

Creatine kinase-MB

- High specificity for cardiac tissues
- Normal range: Adult 0- 5ng/mL
- Begins to rise 4-6 hours after onset of infarction
 - Peaks at about 12 hours
- Returns to baseline at 24-36 hours
 - Can be used to indicate early re-infarction if the level normalizes and then increases again

Myoglobin

- found in cardiac and skeletal muscles.
- single polypeptide with one oxygen binding site.
- It contains a heme prosthetic group that can reversibly bind to oxygen.
- The body uses it as an oxygen storage protein in muscle.
- bind and release oxygen depending on the oxygen conc. in the cell → primary function to supply oxygen to myocytes.
- It is the reason for the red color of the muscle.

Myoglobin cont'd...

- It is sensitive index of myocardial damage.
- Surges often occur earlier than Tn & CK
- It is not specific
- Released from damaged tissues within 1 hour
- Conditions that increase myoglobin
 - ✓ Acute myocardial infarction
 - ✓ Skeletal muscle damage
 - ✓ Renal failure

Troponin

- a complex of three regulatory proteins that is integral to muscle contraction in skeletal & cardiac muscle.
 - Involved in the interaction between actin and myosin for contraction.
- It has three subunits: TnC, TnT & TnI
 - **Troponin-C** has calcium binding ability and has no diagnostic value
 - **Troponin-T** binds the troponin-tropomyosin complex
 - **Troponin-I** is an inhibitory protein

Troponin.....cont'd

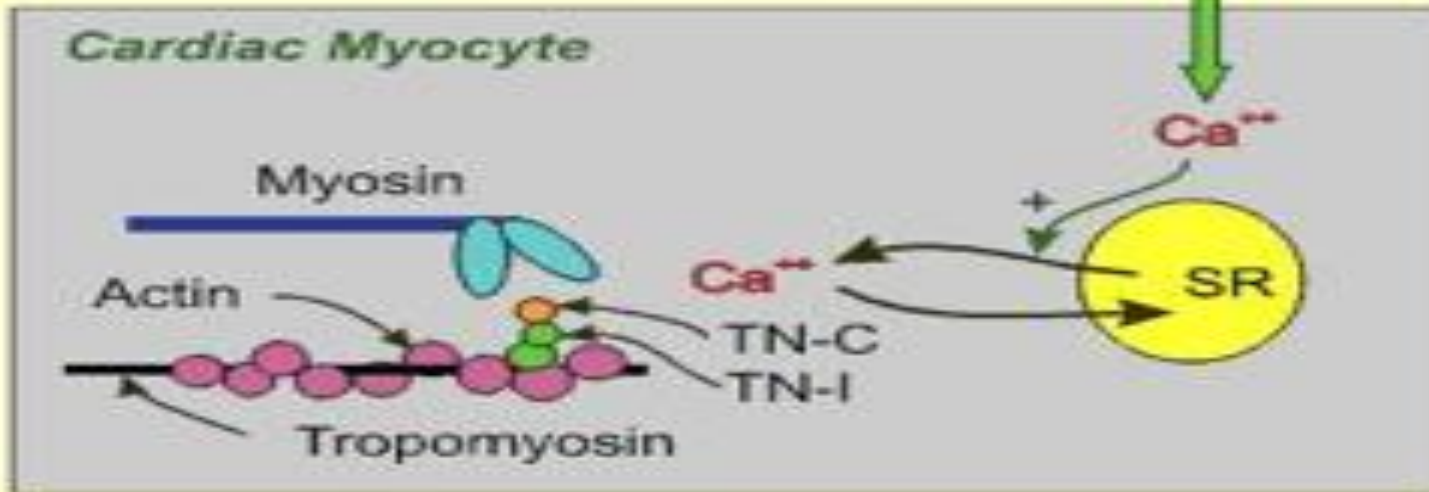


Figure 2. Cardiac myofilaments. Myosin (thick filament) contains two heads having ATPase activity. Thin filament is made up of actin, tropomyosin, and troponin (TN). TN-C binds Ca^{++} released by the sarcoplasmic reticulum (SR). TN-I inhibits actin-myosin binding until Ca^{++} binds to TN-C.

Troponin C

- 2 isoforms (cardiac and slow-twitch muscle)
- Homology between isoforms limits cardiac specificity and diagnostic usefulness
- Binds calcium to initiate muscle contraction
- Calcium binds cTnC → cTnI displaced → tropomyosin conformational change (no longer interferes with myosin/actin) → muscle contraction occurs

Troponin I

- Unique to heart
 - Healthy animals may have low, but detectable levels
- Time to initial elevation = 3-8 hours
- Time to peak elevation = 12-24 hours
- Time to return to baseline = 5-10 days
- Cardiac troponin I (cTnI) is a cardiac muscle protein with a molecular weight of 24 kilo daltons
- cTnI has additional amino acid residues on its N-terminal that are not exist on the skeletal form
- Half life 2-4 hours & serum increase within 2-8 hours
- Troponin I
 - 3 isoforms (2 skeletal, **1 cardiac**)
 - Not expressed in fetal tissue or after damage/regeneration in adult tissue
 - Inhibits actomyosin ATPase, preventing the structural interaction of myosin with actin-binding sites

Troponin T

- Cardiac troponin T (cTnT) is present in fetal skeletal muscle.
- In healthy adult skeletal muscle cTnT is absent.
- Has similar half life and early serum increases as that of cTnI
- Troponin T
 - Multiple isoforms in skeletal tissue
 - 4 isoforms in human cardiac tissue (1 adult, 3 fetal tissue)
 - Fetal isoforms may be re-expressed during heart failure or in damaged skeletal muscle
 - Attaches the troponin complex to tropomyosin and actin

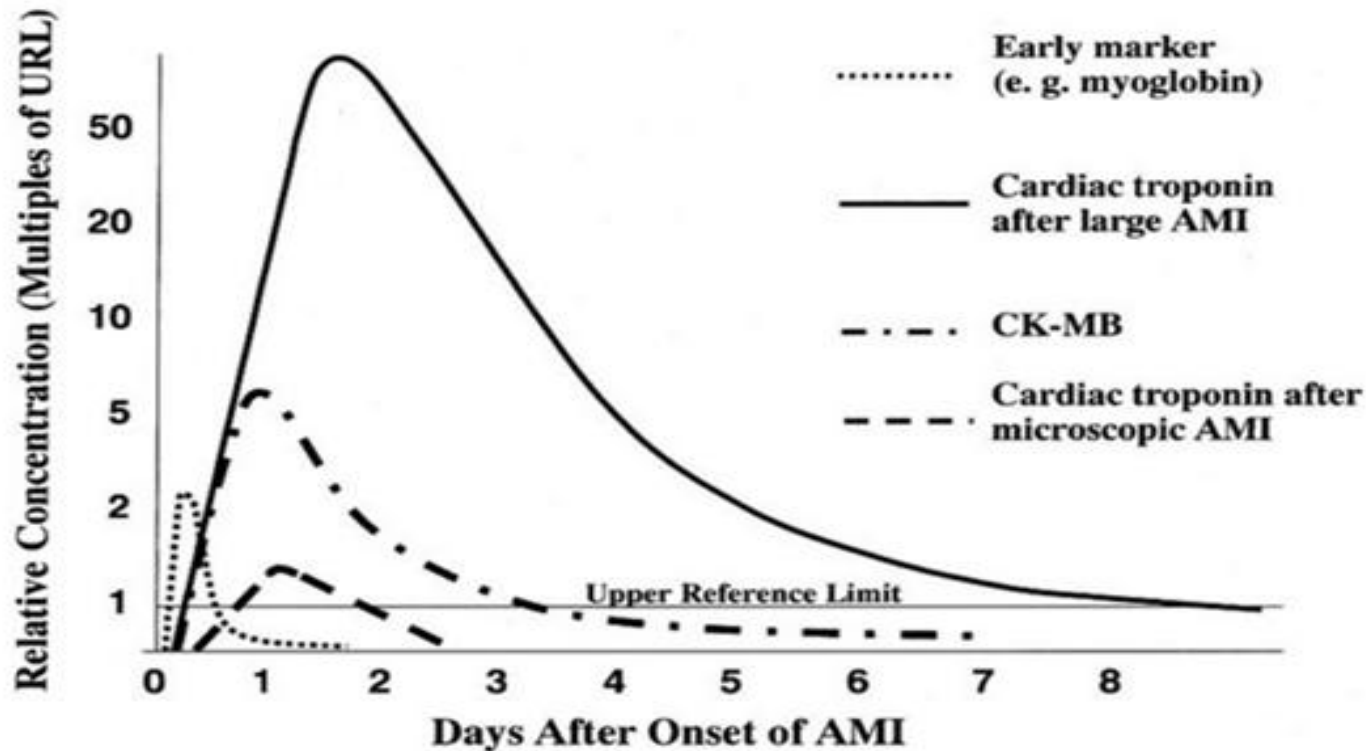
Troponin levels

- Less than 5% in cytosol.
- Troponin levels begin to rise 2-3 hours after onset of myocardial injury
- Elevations of TnI & TnT can persist for up to 10 days after MI.

NB:

- CK-MB returns to baseline by 48 hours

Summary of MI markers

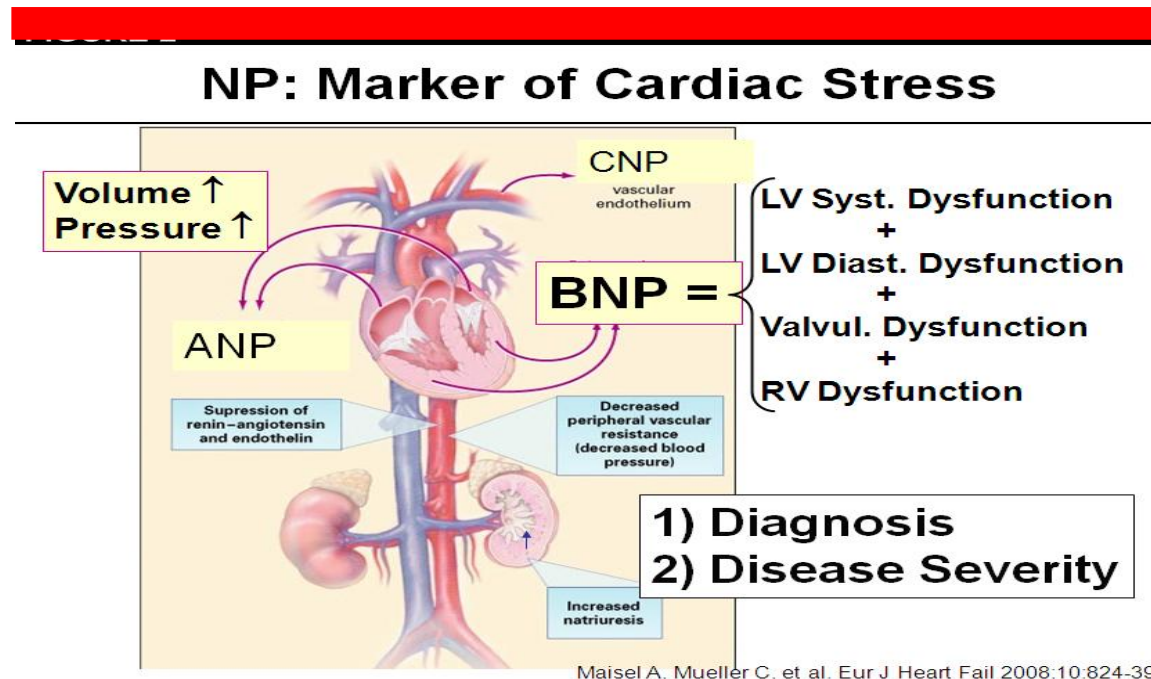


Natriuretic peptides

- NPs are group of structurally similar but genetically distinct neurohormones which acts to maintain CV homeostasis.
- NPs family includes
 - **ANP**: atrial natriuretic peptide (28 aa)
 - predominantly synthesized in the atria and released in response to atrial distension.
 - **BNP**: B-type natriuretic peptide (32 aa)
 - mainly produced and secreted by ventricular myocytes following volume overload → ventricular wall stretch .
 - **CNP**: C-type natriuretic peptide (22 aa)
 - Is found predominantly in the brain and also synthesized by vascular endothelial cells.

Natriuretic peptides.....

- In contrast to ANP, the circulating physiological levels of BNP are generally very low and
 - only become more notable in pathological disease states.



Comparison different families of NPs

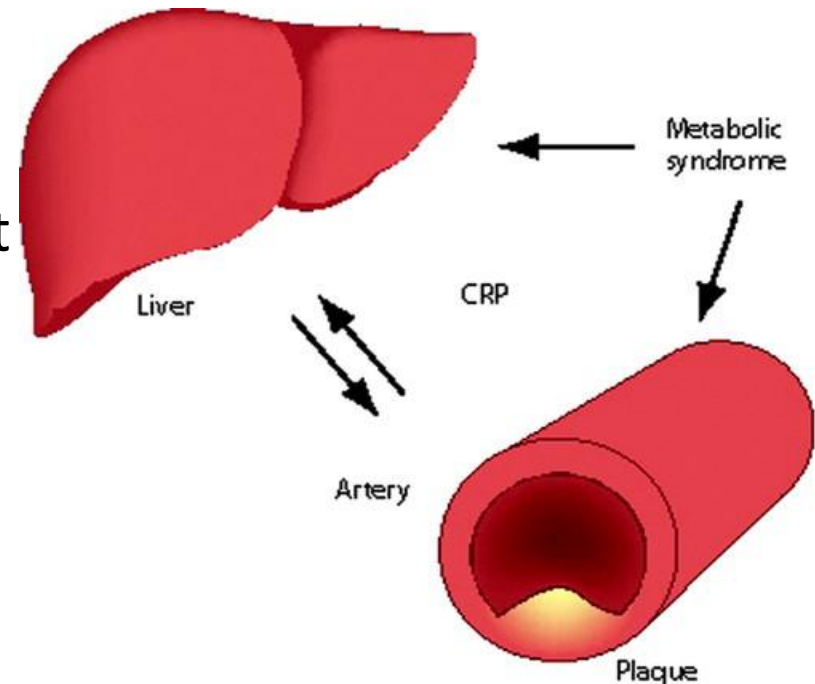
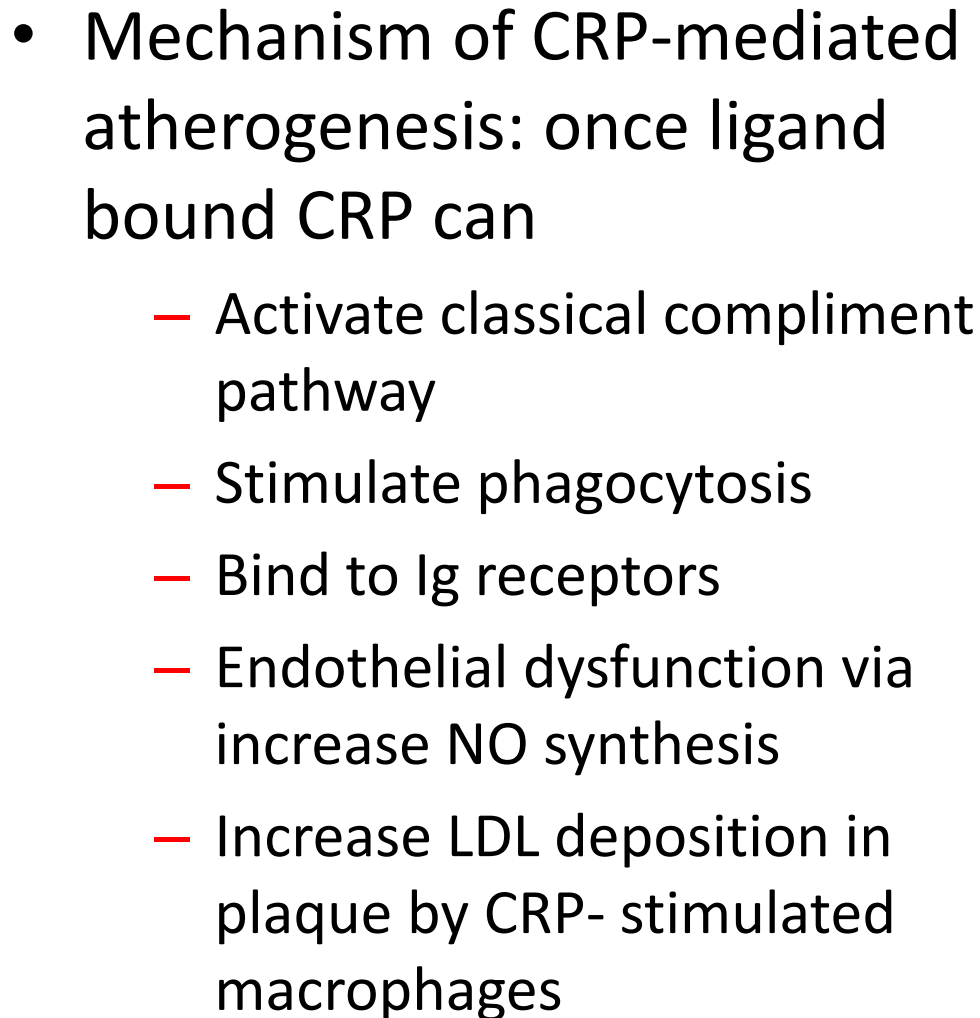


NP	ANP	BNP	CNP
Structure	28 amino acid peptide	32 aa peptide	22 aa peptide
Site of synthesis	Atria	Ventricles	Vascular endothelial cells
Trigger(s)	Atrial distension	Volume overload (→ventr. wall stretch)	Cytokines e.g. IL-1, TNF
Proposed main role	Physiological	Cardiac stress hormone	?
Receptor	NPR-A	NPR-A	NPR-B
Receptor coupling mechanism	↑Guanylate cyclase/cGMP	↑Guanylate cyclase/cGMP	↑Guanylate cyclase/cGMP
Main physiological effects of NP	Natriuresis, diuresis Vasodilation, ↑Renal blood flow & GFR ↑Myocardial relaxation	Natriuresis, diuresis Vasodilation, ↑Renal blood flow & GFR ↑Myocardial relaxation Lipid mobilization,	Vasodilation Antihypertrophic and antifibrotic Anti-inflammatory Antithrombotic

C-reactive protein

- CRP is an acute-phase protein produced by the liver
- Pentameric structure consisting of five 23 kDa identical subunits
- Plasma levels can increase rapidly to 10,000x levels
- CRP useful marker for increased cardiovascular risk.
- This is an attractive concept because atherosclerosis is a disease characterized by chronic arterial inflammation and
 - suggests the possibility that subclinical states of atherosclerosis.

C-reactive protein.....



Less Commonly Used Cardiac Markers

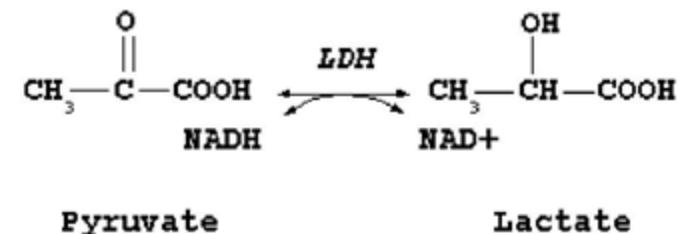
❖ Lactate dehydrogenase system

- ✓ were used widely in the past for diagnosis of myocardial infarction
- ✓ exists in five isoenzyme forms

LDH ISOENZYMES

LDH exists in 5 forms (isoenzymes), which differ slightly in structure. • All of these isoenzymes can be measured in the blood, and can be separated by electrophoresis.

LDH Isoenzyme	Tissues
LDH-1	is found primarily in heart muscle and red blood cells.
LDH-2	is concentrated in white blood cells.
LDH-3	is highest in the lung
LDH-4	is highest in the kidney, placenta, and pancreas
LDH-5	is highest in the liver and in skeletal muscle





Lactate dehydrogenase system.....

- In particular, these enzymes and isoenzymes carry out their metabolic functions within specific cells, and are released into body fluids when those cells become diseased.**
- Thus an increase in LD isoenzyme activity can indicate leakage from cell due to cellular injury.**
- The level of isoenzyme LD1 compared with LD2 has been used to detect an AMI because of the high concentration of LD in cardiac muscle fibres.**
- LD isoenzymes begin to leak out of dying heart muscle cells and are detectable in the serum by 12 to 24 hours following a heart attack**

Lipid Profile Values

❖ Standard fasting blood tests for cholesterol and lipid profiles

will include values for:-

- Total cholesterol,
- HDL cholesterol
- LDL cholesterol
- Triglycerides.

Lipid Profile Values

❖ values for various lipids that indicate low to high risk for coronary artery disease.

1. Total Serum Cholesterol

- ✓ <200mg/dL = desired values
- ✓ 200–239mg/dL = borderline to high risk
- ✓ 240mg/dL and above = high risk

Lipid Profile Values

II. HDL Cholesterol: With HDL cholesterol the higher the better

➤ <40mg/dL for men &

➤ <50mg/dL for women

Higher risk

➤ 40–50mg/dL for men &

➤ 50–60mg/dL for woman

Normal values

➤ >60mg/dL:-

- *is associated with some level of protection against heart disease*

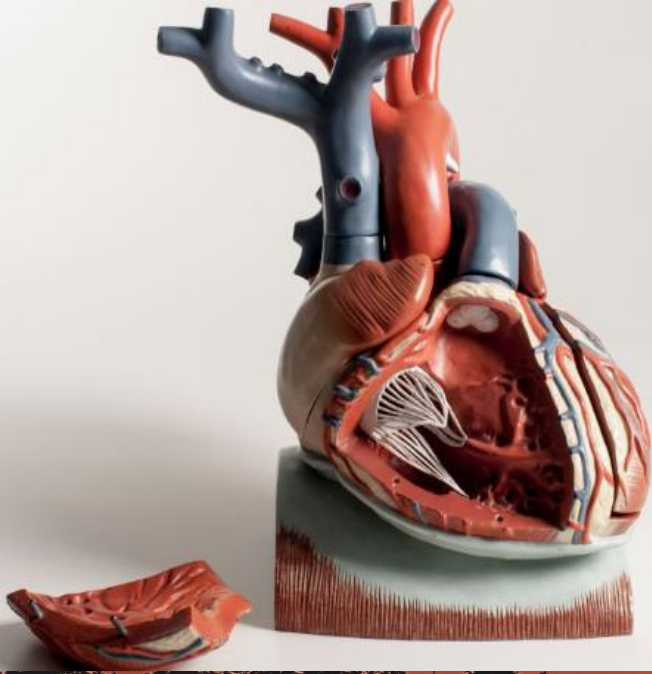
Lipid Profile Values

III. LDL Cholesterol :- With LDL cholesterol the lower the better

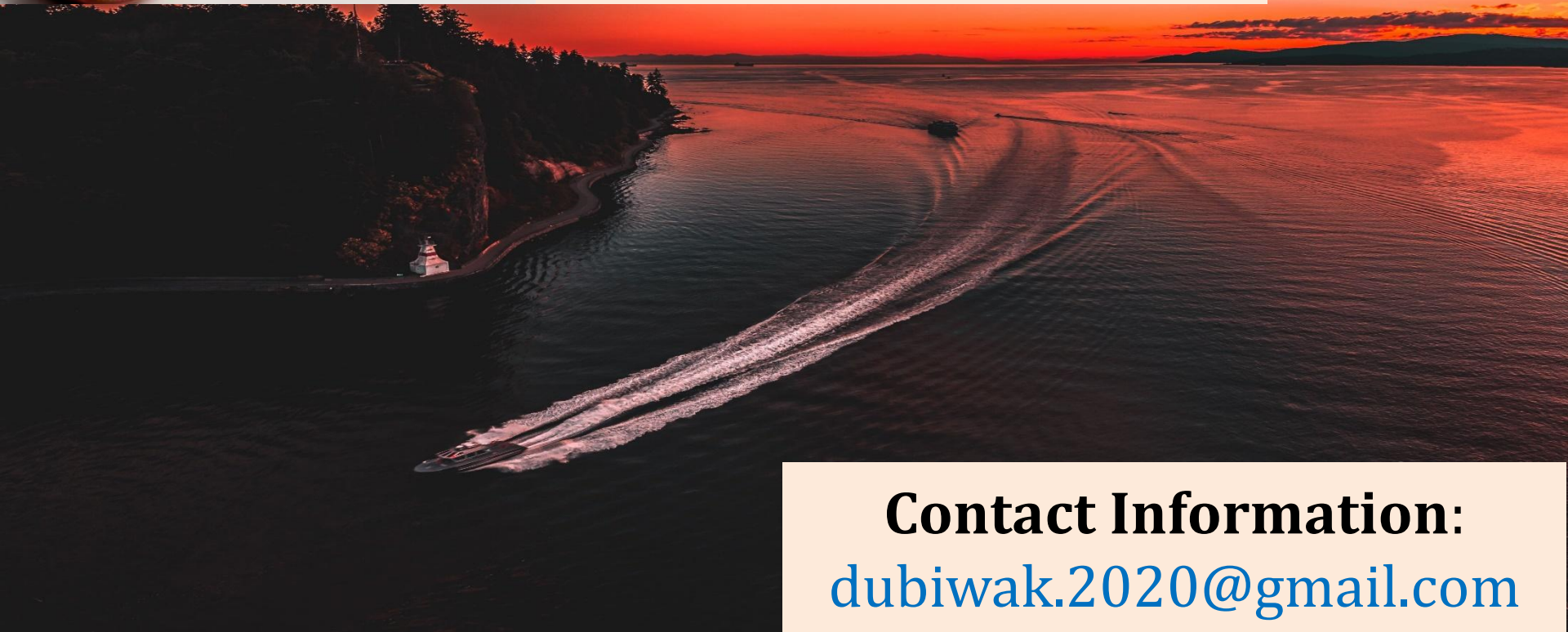
- $<100\text{mg/dL}$ = **optimal values**
- $100\text{mg/dL}-129\text{mg/dL}$ = **optimal to near optimal**
- $130\text{mg/dL}-159\text{mg/dL}$ = **borderline high risk**
- $160\text{mg/dL}-189\text{mg/dL}$ = **high risk**
- 190mg/dL and higher = **very high risk**

IV. Triglycerides:- With triglycerides the lower the better

- ✓ $<150\text{mg/dL}$ = **normal**
- ✓ $150\text{mg/dL}-199\text{mg/dL}$ = **borderline to high risk**
- ✓ $200\text{mg/dL}-499\text{mg/dL}$ = **high risk**
- ✓ $>500\text{mg/dL}$ = **very high risk**



THANK YOU!!!



Contact Information:
dubiwak.2020@gmail.com